English

(0000CJA101021230016)

Test Pattern

ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
AIOT
21-04-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

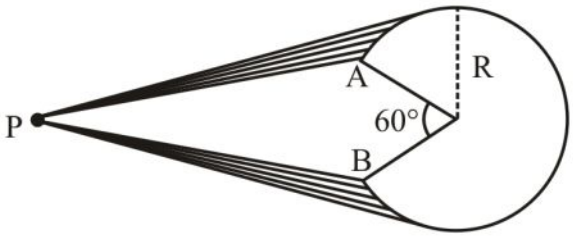
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

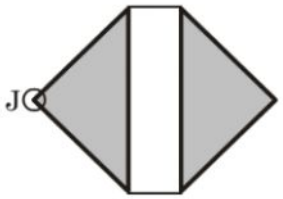
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1. A nonconducting hollow cone has charge density $12\varepsilon_0$ C/m^2. A part ABP is cut and removed from the cone. The radius of the base of the cone is 1m. The potential due to the remaining portion of the cone at point 'P' is



(A) 5V (B) 6V (C) 8V (D) 10V

2. A square of side length a and mass 2m is cut across its diagonal. Two ideal light rods of negligible length are rigidly attached to the two corners that were cut. A tangential impulse J (into the page) is applied to the outermost corner of the square. What is the tension developed in each rod?



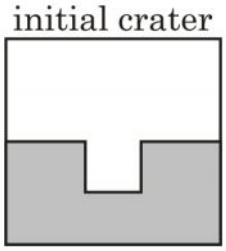
(A) $\dfrac{9(2-\sqrt{2})J^2}{ma}$

(B) $\dfrac{27(2-\sqrt{2})J^2}{ma}$

(C) $\dfrac{3J^2}{\sqrt{2}ma}$

(D) $\dfrac{3J^2}{2\sqrt{2}ma}$

3. A meteor hits the moon and creates a crater within the ground. The crater can be modelled as a cubic hole of length L, created when the material originally within it is ejected out after the collision. Assume that energy is not lost during the collision, and kinetic energy of meteor gets converted to gravitational potential energy of the ejected mass. If a meteor of twice the mass of the initial meteor hits the moon at the same velocity, how much larger is the volume of the crater compared to the initial one? Assume everything on the moon are of the same density.



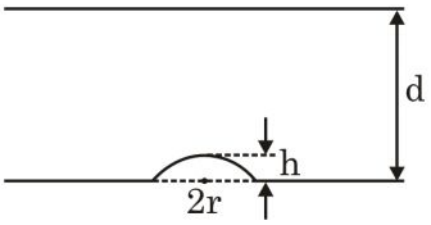
(A) 16

(B) 4

(C) 2

(D) $2^{3/4}$

4. On one of the plates of the capacitors has a small hole of radius r, covered with a soap film. The surface tension of the film is σ. The capacitor is charged to a potential V. Distance d between the plates is small compared to linear dimensions of the plates. Value of h (<< r) is equal to :-



(A) $\frac{\varepsilon_0 V^2 r^2}{8\sigma d^2}$

(B) $\frac{\varepsilon_0 V^2 r^2}{4\sigma d^2}$

(C) $\frac{\varepsilon_0 V^2 r^2}{16\sigma d^2}$

(D) $\frac{\varepsilon_0 V^2 r^2}{\sigma d^2}$

SECTION-I (ii) : (Maximum Marks: 12)

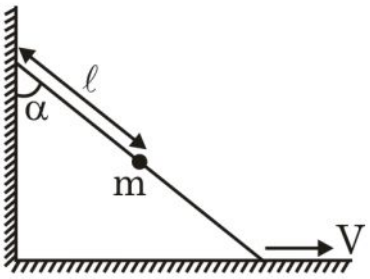
- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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5. The length of a weightless rod is 2ℓ. A small sphere of mass m is fixed at a distance ℓ from its upper end. The rod rests with its one end against the wall and the other against the floor. The end that rests on the floor is being moved with a constant velocity V away from the wall :-



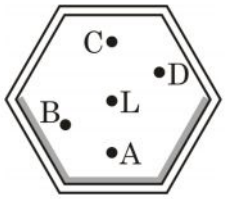
(A) The force with which the sphere affects the rod at the moment when $\alpha = 45°$ is $m\left(g - \frac{V^2}{\sqrt{2}\,\ell}\right)$.

(B) The force with which the sphere affects the rod at the moment when $\alpha = 45°$ is $m\left(g - \frac{V^2}{2\sqrt{2}\,\ell}\right)$.

(C) Vertical acceleration of m when $\alpha = 60°$ is $\frac{2V^2}{\ell}$.

(D) Horizontal acceleration of m when $\alpha = 60°$ is $\frac{V^2}{\ell}$.

6. Three adjacent walls of a regular hexagon-shaped room are covered with flat mirror (see the figure). A lamp is lit at the centre of the room.



(A) The number of images seen by an observer at A is 4.

(B) The number of images seen by an observer at B is 4.

(C) The number of images seen by an observer at C is 5.

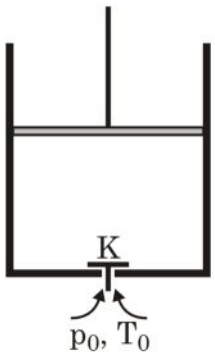
(D) The number of images seen by an observer at D is 5.

7. At the end of the thermally insulated cylinder with the piston there is a valve (figure) that covers a small hole, which opens and begins to let air from outside into the cylinder at a pressure difference $\Delta P = P_0/3$ (P_0 is atmospheric pressure). The valve does not allow air to escape from the cylinder to the outside. At the initial moment of time, the piston is pressed against the end of the cylinder, there is no air inside.

In the first case, the cylinder is filled with air to volume V_0, slowly moving the piston, after which it is stopped, and then the piston is released.

In the second case, the piston is abruptly moved to a position in which the volume under the piston is equal to V_0, so that the air does not have time to penetrate through the valve into the cylinder. In this position, fix the piston, wait until the cylinder is filled with air, just as in the first case, release the piston.

Out side the cylinder is air at atmospheric pressure P_0 and temperature T_0. The friction of the piston against the walls, the mass of the piston, as well as the heat exchange of air with the piston and the cylinder walls can be neglected. Air can be considered a diatomic ideal gas. After the piston is released, the valve remains closed all the time.



(A) For first case the temperature of air inside the cylinder at the moment the piston is stopped at volume V_0 is T_0.

(B) For first case, the temperature of air inside the cylinder when the piston stops movement is $\frac{8}{7}T_0$.

(C) For second case, the temperature of air in the cylinder after stopping of piston and filling with air is $\frac{7}{5}T_0$.

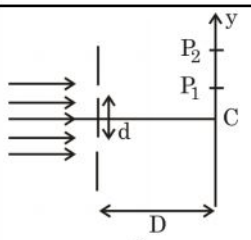
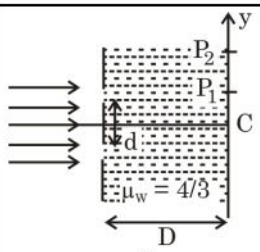
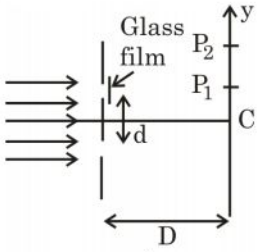
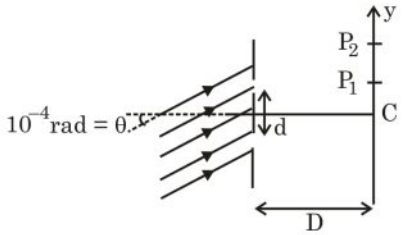
(D) For second case, the temperature of air after piston is released and it stops movement is $\frac{9}{5}T_0$.

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

8. List–I contains four different YDSE systems and List–II contains intensity measured at some points on the screen. Do the correct match (es) in List–I and List–II, ($CP_1 = 0.3$ mm and $CP_2 = 1.2$ mm)

	List–I		List–II
(P)	 $\lambda = 4000$ Å in air d = 1mm, D = 1m Intensity due to each slit = I_0	(1)	$I_C = 4I_0$
(Q)	 $\lambda = 4000$ Å in air d = 1mm, D = 1m Intensity due to each slit = I_0 and no absorption of light by water ($\mu_W = 4/3$)	(2)	$I_{P_1} = 2I_0$
(R)	 $\lambda = 4000$ Å in air d = 1mm, D = 1m Film thickness t = 0.8 μm, refractive index of film = 3/2 Intensity due to each slit = I_0 and no absorption of light by glass	(3)	$I_{P_1} = 4I_0$
(S)	 $\lambda = 4000$ Å in air d = 1mm, D = 1m Intensity due to each slit = I_0	(4)	$I_{P_1} = 0$
		(5)	$I_{P_2} = 2I_0$

(A) P ⟶ 1,2;Q ⟶ 1,3;R ⟶ 2;S ⟶ 4,5

(B) P ⟶ 1,2;Q ⟶ 1,3;R ⟶ 2,5;S ⟶ 4,5

(C) P ⟶ 2;Q ⟶ 3;R ⟶ 2,5;S ⟶ 1,2

(D) P ⟶ 1,2;Q ⟶ 1,3;R ⟶ 1,2;S ⟶ 4,5

9. An electron in a hydrogen atom makes a transition $n_1 \longrightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of the two states. Assume Bohr model to be valid.

	List–I		List–II
(P)	The electron emits an energy of 2.55 eV	(1)	$n_1 = 2, n_2 = 1$
(Q)	Time period of the electron in the initial state is eight times that in the final state.	(2)	$n_1 = 4, n_2 = 2$
(R)	Speed of electron become two times	(3)	$n_1 = 5, n_2 = 3$
(S)	Radius of orbit of electron becomes 4.77 Å	(4)	$n_1 = 6, n_2 = 3$
		(5)	$n_1 = 8, n_2 = 4$

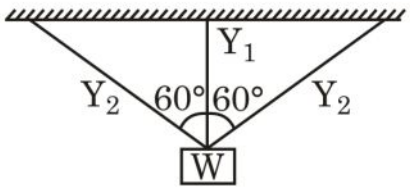
(A) P $\longrightarrow$ 2;Q $\longrightarrow$ 1,2;R $\longrightarrow$ 1,2;S $\longrightarrow$ 3

(B) P $\longrightarrow$ 2;Q $\longrightarrow$ 1,2,4;R $\longrightarrow$ 1,3,4;S $\longrightarrow$ 4

(C) P $\longrightarrow$ 2;Q $\longrightarrow$ 1,2,4,5;R $\longrightarrow$ 1,2,4,5;S $\longrightarrow$ 3,4

(D) P $\longrightarrow$ 2;Q $\longrightarrow$ 1,2,3,4,5;R $\longrightarrow$ 1,2,3,4,5;S $\longrightarrow$ 4

10. A load is supported using three wires of same cross section area as shown.



	List–I		List–II
(P)	Equal tensile stress in all wires	(1)	$Y_2 = 2Y_1$
(Q)	Equal tension in all wires	(2)	$Y_2 = 4Y_1$
(R)	Equal elastic potential energy in all wires	(3)	$Y_2 = 8Y_1$
(S)	Equal elastic potential energy per unit volume in all wires	(4)	$Y_2 = 16Y_1$
		(5)	Does not depend on ratio of $\frac{Y_1}{Y_2}$

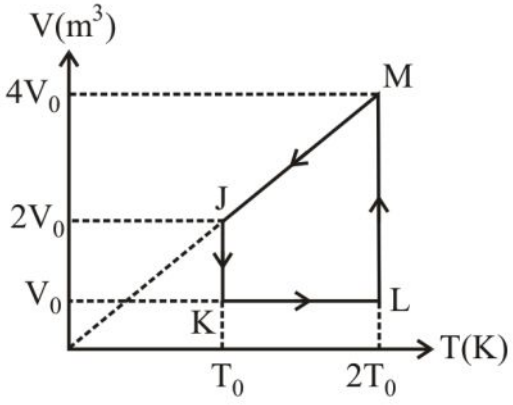
(A) P $\longrightarrow$ 2;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(B) P $\longrightarrow$ 2;Q $\longrightarrow$ 3;R $\longrightarrow$ 2;S $\longrightarrow$ 4

(C) P $\longrightarrow$ 4;Q $\longrightarrow$ 5;R $\longrightarrow$ 3;S $\longrightarrow$ 1

(D) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 5;S $\longrightarrow$ 4

11. V–T graph of a thermodynamic process is shown in figure. If heat given to process is taken to be positive :- (ΔQ is heat given to the gas in the process, ΔU is change in internal energy in the process, ΔW is the work done by the gas in the process).



	List–I		List–II
(P)	JK	(1)	$\Delta W \geq 0$
(Q)	KL	(2)	$\Delta Q < 0$
(R)	LM	(3)	$\Delta W < 0$
(S)	MJ	(4)	$\Delta Q > 0$
		(5)	$\Delta U = 0$

(A) P $\rightarrow$ 2,3,5;Q $\rightarrow$ 4;R $\rightarrow$ 1,4,5;S $\rightarrow$ 2,3

(B) P $\rightarrow$ 1,3,5;Q $\rightarrow$ 4;R $\rightarrow$ 1,4,5;S $\rightarrow$ 2,5

(C) P $\rightarrow$ 2,3,5;Q $\rightarrow$ 1,4;R $\rightarrow$ 1,4,5;S $\rightarrow$ 2,3

(D) P $\rightarrow$ 1,4;Q $\rightarrow$ 2,4;R $\rightarrow$ 1,3,5;S $\rightarrow$ 2,3

SECTION-II : (Maximum Marks: 24)

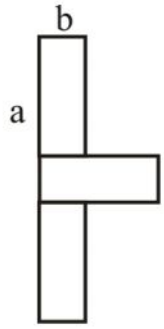
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases.

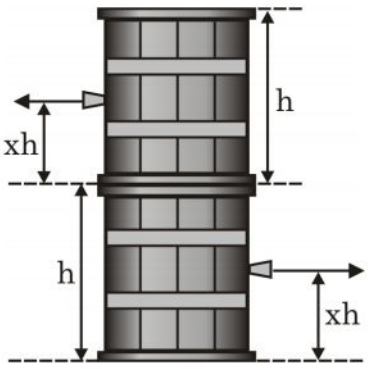
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1. The pitch of a screw gauge is 1 mm and its cap is divided into 100 divisions. When nothing is placed between its studs, the zero of the circular scale lies 6 divisions below the reference line and zero of main scale is just visible. When a wire is placed between its studs, the main scale reading is 4 mm and 56th division of circular scale coincides with the reference line. The length of the wire is 4.20 cm. If the curved surface area (in cm^2) comes out to be x, then find the value of 100x. $\left(\pi = \frac{22}{7}\right)$

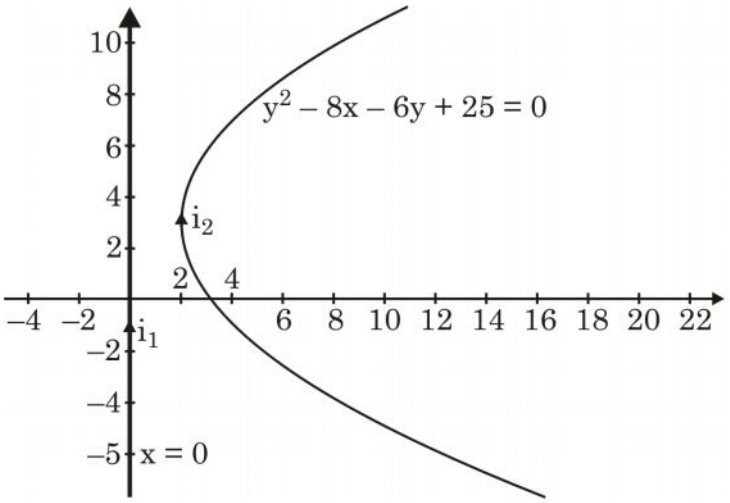
2. A tower is glued together from three identical wooden blocks of edge lengths a and b as in the figure below. What is the largest ratio a/b such that the tower would still be stable?



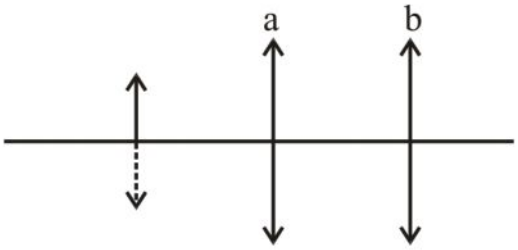
3. I placed 2 identical cylinder-shaped barrels, having the same height h, on the horizontal ground, one on the top of the other (see the figure). Both were nearly fully filled with water. Your task is to drill a hole perpendicularly to the wall of each of the barrels at the same xh height measured from the bottom of each barrel. The value of the dimensionless factor x, in order that the impact points of the water rays fall as far apart as possible. The value of $\frac{1}{x}$ is :-



4. Two infinitely long current carrying wires carry constant current i_1 and i_2 as shown in the diagram. The equations of the wire curvatures are $y^2 - 8x - 6y + 25 = 0$ and $x = 0$. The magnitude of force acting on one of the wires due to the other is $\frac{\mu_0 i_1 i_2}{\beta}$. The value of β is (The current-carrying wires are rigidly fixed. The units for distances on the graph should be taken in meters.)



5. Consider two convex lens with focal length f_a and f_b respectively. If placed appropriately, a small object perpendicular to the optic axis can create an inverted image, the size and position of which matches the size and position of object as shown in the figure. The distance between the lens is d and distance between the object and the left lens is u. The ratio $\frac{u}{d}$ is (Take : $f_a = 1$ cm, $f_b = 4$ cm)



6. A capacitor of value 1.0 F discharges through a device whose resistance R varies linearly with applied potential difference, V, so that $R = AV + B$, where A and B are constants. The resistance of the device has a value of 10.0Ω when V = 6.0 V, and 4.0Ω when V = 0.06 V. The capacitor is initially charged to a potential of 6.0V. Determine how long (in sec) takes for the capacitor to discharge to 1% of this initial value. (Take : $\ell n10 = 2$)

PART-2 : CHEMISTRY

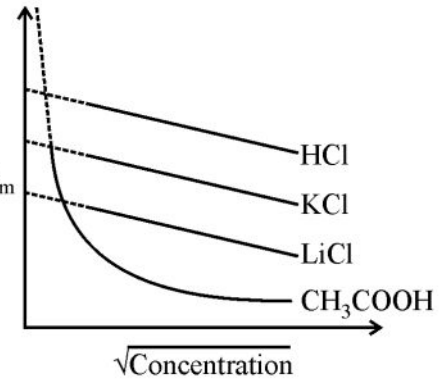
SECTION-I (i) : (Maximum Marks: 12)

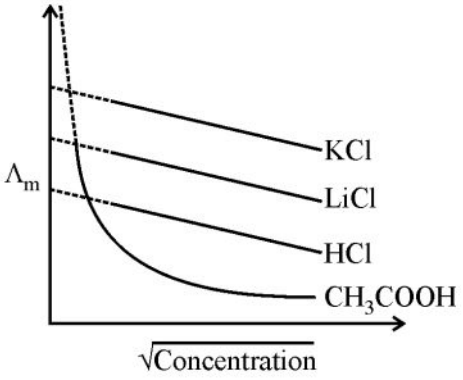
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

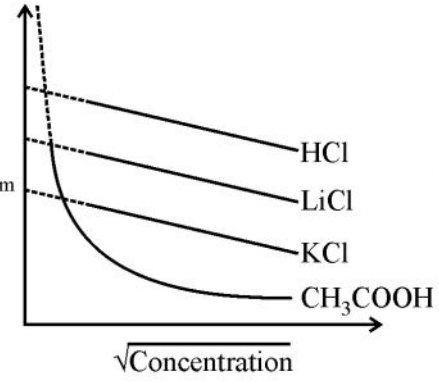
Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

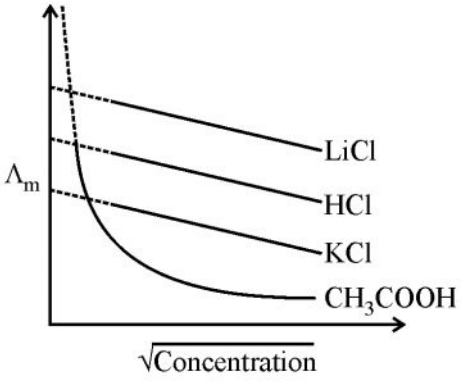
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1. Select the correct option for following aqueous solutions :

(A) 

(B) 

(C) 

(D) 

2. $H_3C - C \equiv C - Ph \xrightarrow{Hg^{2+}/H^+/H_2O} P$, P is :

(A) $H_3C - \underset{\underset{O}{\|}}{C} - CH_2 - Ph$

(B) $H_3C - CH_2 - \underset{\underset{O}{\|}}{C} - Ph$

(C) $H_3C - \underset{\underset{O}{\|}}{C} - \underset{\underset{O}{\|}}{C} - Ph$

(D) $H_3C - \underset{\underset{OH}{|}}{CH} - CH_2 - Ph$

3. Alkaline solution of (X) $\xrightarrow{NaBO_3}$ Canary yellow colored solution

Acidic solution of (Y) $\xrightarrow{NaBiO_3}$ Lilac colored solution

(X) & (Y) are respectively :-

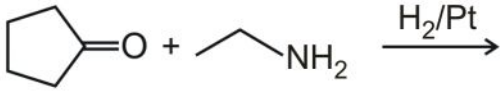
(A) $Fe(OH)_3$, $Mn(OH)_2$

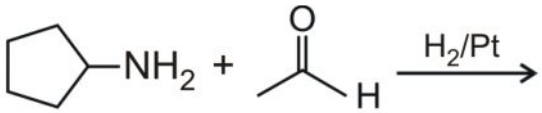
(B) $Cr(OH)_3$, $Pb(OH)_2$

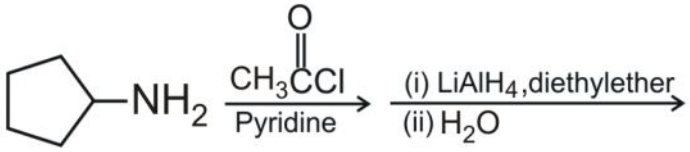
(C) $Fe(OH)_3$, $Zn(OH)_2$

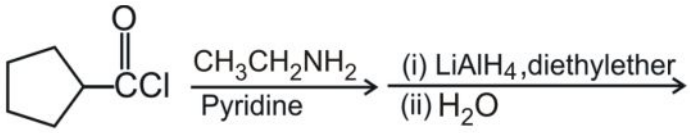
(D) $Cr(OH)_3$, $Mn(OH)_2$

4. Among the following, which reaction will not give N-ethylcyclopentylamine as major product?

(A) 

(B) 

(C) 

(D) 

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Consider two liquids A and B having pure vapour pressures P_A^o & P_B^o forming an ideal solution. The plot of $\frac{1}{X_A}$ v/s $\frac{1}{Y_A}$ (where X_A and Y_A are the mole fraction of liquid A in liquid and vapour phase respectively) is linear with slope and Y intercepts respectively :

(A) Intercept = $\frac{P_A^o - P_B^o}{P_B^o}$

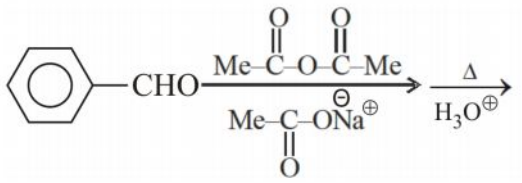
(B) Intercept = $\frac{P_B^o - P_A^o}{P_B^o}$

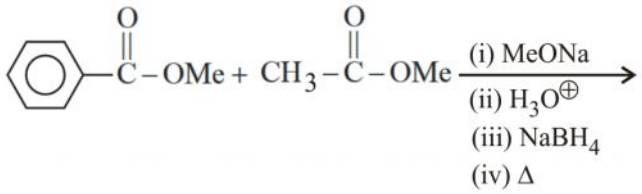
(C) Slope = $\frac{P_A^o}{P_B^o}$

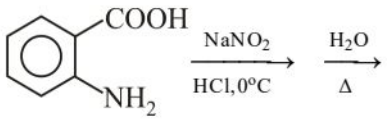
(D) Slope = $\frac{P_B^o}{P_A^o}$

6. In which of the following reactions cinnamic acid is one of the major organic product ?

(A) $Ph-CH_2-COOEt \xrightarrow[(ii)H_3O^{\oplus}]{(i)Et\bar{O}Na^{\oplus}} \xrightarrow[(iii)\Delta]{}$

(B) 

(C) 

(D) 

7. The complex(es) with distorted octahedron structure is/are :-

(A) $[Cu(H_2O)_6]^{2+}$

(B) $[FeF_6]^{3-}$

(C) $[MnF_6]^{3-}$

(D) $[Fe(CN)_6]^{4-}$

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

---

8. Four different gases and their Vander Waal's constant are given below : (R = 0.08 L-atm/K-mol)

Gas	a (in atm L^2 / mol^2)	b (in L/mol)
W	1.5	0.05
X	2.0	0.08
Y	4.2	0.025
Z	5.0	0.10

List - I		List - II	
(P)	W	(1)	Gas having highest intermolecular force of attraction.
(Q)	X	(2)	Gas having smallest size of molecules.
(R)	Y	(3)	Gas that can be liquefied at 85 K
(S)	Z	(4)	Gas resembles with ideal gas at nearly boiling point of water
		(5)	Gas with highest boyle's temperature.

Codes :

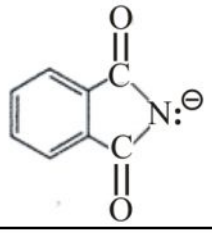
(A) P $\longrightarrow$ 3,4,5;Q $\longrightarrow$ 1,5;R $\longrightarrow$ 2,4;S $\longrightarrow$ 5

(B) P $\longrightarrow$ 1,2;Q $\longrightarrow$ 4,5;R $\longrightarrow$ 3,4;S $\longrightarrow$ 2

(C) P $\longrightarrow$ 4;Q $\longrightarrow$ 3;R $\longrightarrow$ 2,5;S $\longrightarrow$ 1

(D) P $\longrightarrow$ 1,4;Q $\longrightarrow$ 3,4;R $\longrightarrow$ 5;S $\longrightarrow$ 1,2,3

9. For the reaction, using suitable alkyl halide and a suitable nucleophile what will be the correct match for the nucleophile used as per List-I and the solvent being used in List-II?

List-I		List-II	
(P)	MeS^-	(1)	Reacts with CH_3CH_2Cl to give SN^2 product
(Q)	$NC^- + NH_4Cl$	(2)	Reacts with CH_3CHO followed by hydrolysis to give alanine.
(R)		(3)	Reacts with EtX followed by NH_2NH_2gives $EtNH_2$
(S)	$Ph_3P: + nBuLi$	(4)	Reacts with CH_3CH_2Cl followed by acetone to give 2-methyl-but-2-ene.
		(5)	Reacts with chloro benzene followed by H_3O^+ give aniline.

(A) P ⟶ 1;Q ⟶ 2;R ⟶ 3;S ⟶ 4

(B) P ⟶ 5;Q ⟶ 2;R ⟶ 1;S ⟶ 4

(C) P ⟶ 5;Q ⟶ 4;R ⟶ 3;S ⟶ 1

(D) P ⟶ 1;Q ⟶ 4;R ⟶ 5;S ⟶ 3

10.

List-I (Molecules/ions)		List-II (Arrangement of given entity in List-I)	
(P)	CO on each manganese in $[Mn_2(CO)_{10}]$	(1)	Triangular pyramidal
(Q)	Cl on each Al in Al_2Cl_6	(2)	Octahedral
(R)	Two S around each S in S_8 molecule	(3)	Square pyramidal
(S)	Oxygen atoms around each sulphur in $S_2O_6^{2-}$	(4)	Tetrahedral
		(5)	Bent

Select **CORRECT** code :-

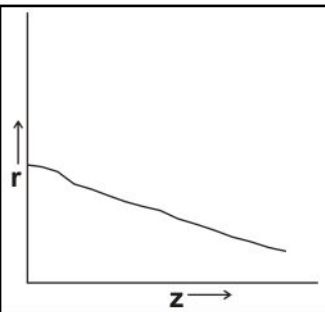
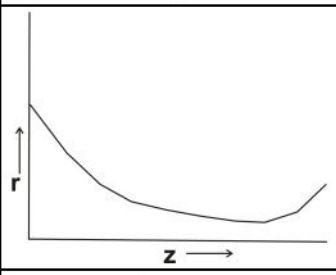
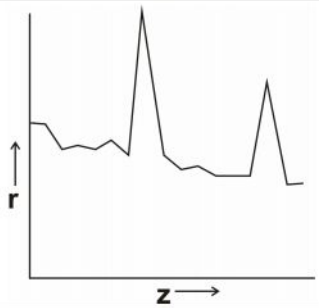
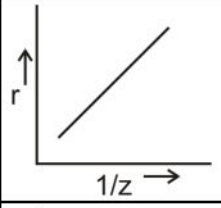
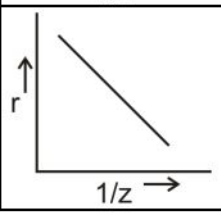
(A) P ⟶ 2;Q ⟶ 4;R ⟶ 5;S ⟶ 1

(B) P ⟶ 3;Q ⟶ 4;R ⟶ 5;S ⟶ 1

(C) P ⟶ 3;Q ⟶ 1;R ⟶ 2;S ⟶ 4

(D) P ⟶ 2;Q ⟶ 2;R ⟶ 4;S ⟶ 5

11.

List-I (Element in periodic table)		List-II [Graphical change in radius (r) with atomic number (Z)]	
(P)	Halogens	(1)	
(Q)	M^{+3} ions of Lanthanides	(2)	
(R)	M atoms of Lanthanides	(3)	
(S)	Size of 3d series metal (covalent radius)	(4)	
		(5)	

Select **CORRECT** code :-

(A) P ⟶ 4;Q ⟶ 5;R ⟶ 1;S ⟶ 2

(B) P ⟶ 5;Q ⟶ 1;R ⟶ 3;S ⟶ 2

(C) P ⟶ 4;Q ⟶ 3;R ⟶ 1;S ⟶ 2

(D) P ⟶ 5;Q ⟶ 1;R ⟶ 2;S ⟶ 3

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases.

---

1. Identify correct number of statements :

Statement - 1 : Entropy of system increases in dissociation of ammonia gas into N_2 and H_2 gases.

Statement - 2 : Entropy of system increases in polymerization process of ethene to form polythene.

Statement - 3 : Entropy of system increases in adiabatic irreversible compression of an ideal gas.

Statement - 4 : Entropy of system increases in condensation of H_2O (vapour).

Statement - 5 : Entropy of system increases in isothermal free expansion of ideal gas.

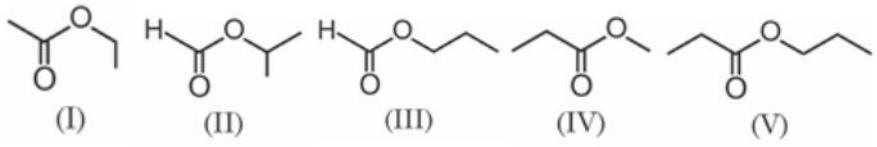
Statement - 6 : Entropy of system always increases in a spontaneous process.

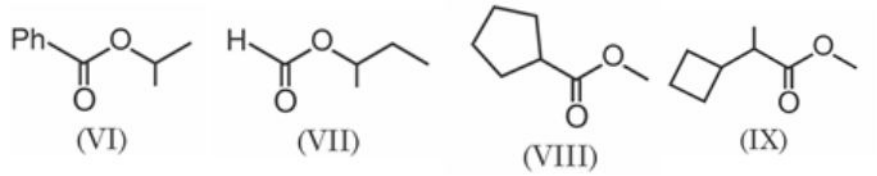
Statement - 7 : Adiabatic process whether reversible or irreversible will not cause any change in entropy of the surroundings.

Statement - 8 : At absolute zero, entropy of any substance becomes zero.

2. 1.2 gram of carbon is burnt completely with 2.56 gram of oxygen to produce CO(g) and CO_2(g). None of the reactant remaining after reaction. Produced mixture of gases is treated with solid I_2O_5 (to know the amount of CO(g) produced). The liberated iodine required x mL of 0.1 M hypo solution for complete titration. The value of x is _____

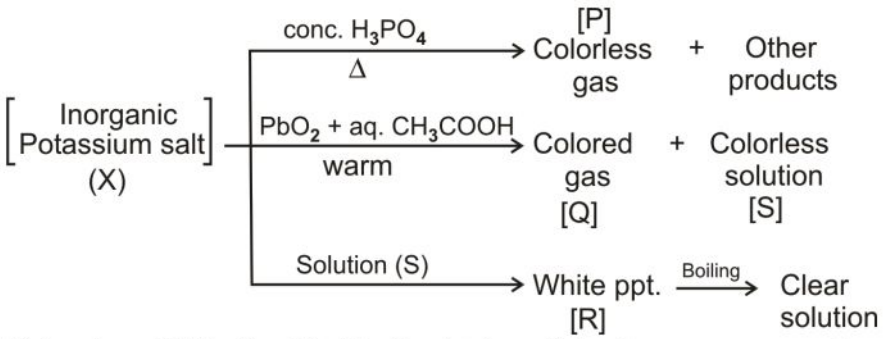
3. An organic compound reacts with excess MeMgBr followed by H_2O to give two organic product of which one reacts with NaOI to give yellow precipitate and OTHER DOES NOT. How many of the following compounds can satisfy this condition ?


(I) (II) (III) (IV) (V)


(VI) (VII) (VIII) (IX)

4. Consider the following reactions and write molecular weight of **white precipitate (Y)**.

(Given atomic mass : H = 1, O = 16, C = 12, F = 19, N = 14, S = 32, P = 31, Cl = 35.5, Br = 80, I = 127, K = 39, Na = 23, Pb = 207, Al = 27, Si = 28, Mn = 55, Cr = 52)



Colored gas [Q] is identified by its staining of starch paper orange red

Phenol + [Q] in water ⟶ **White precipitate (Y)** + other product

5. Find the Mo—Mo bond order in $[(\eta^5\text{-}C_5H_5)Mo(CO)_2]_2$

(Given that both the Mo have 18 valence electrons in given complex)

[Atomic number of Mo = 42]

6. Consider all possible isomeric ketones including stereoisomers of MW = 86, All these isomers are independently reacted with $NaBH_4$ (NOTE : stereoisomers are also reacted separately).

The total number of ketones that give a racemic product(s) is/are.

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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1. If $I = \int_{\frac{1}{4}}^{4} \frac{\tan^{-1}x}{x}dx$, then the value of 2[I] is equal to - (where [.] denotes greatest integer function and $e^{0.693} \simeq 2$)

(A) 6 (B) 4

(C) 2 (D) 8

2. If $\int_{0}^{1} x\, f(x^2)\, dx = \frac{1}{2}$, then $\int_{1}^{2} f^{-1}(x)\, dx$ is equal to (where f(0) = 1, f(1) = 2 & f is strictly increasing in its domain)

(A) 0 (B) 2

(C) 1 (D) 3

3. Number of solutions of $\sqrt{\tan\theta} = 2\sin\theta$, $\theta \in [0, 2\pi]$ is equal to

(A) 2 (B) 4

(C) 5 (D) 7

4. Value of $\lim_{n\to\infty} n \int_{0^+}^{\pi/2} \left(1 - \sqrt[n]{\sin x}\right) dx$ is

(A) $\ell n\, 2$

(B) $-\ell n\, 2$

(C) $\frac{\pi}{2}\ell n\, 2$

(D) $-\frac{\pi}{2}\ell n\, 2$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. If A_i is the area bounded by $|x - a_i| + |y| = b_i$, $i \in N$, where $a_{i+1} = a_i + \frac{3}{2}b_i$ & $b_{i+1} = \frac{b_i}{2}$, $a_1 = 0$, $b_1 = 32$, then

(A) $A_3 = 128$

(B) $A_3 = 256$

(C) $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} A_i = \frac{8}{3}(32)^2$

(D) $\lim\limits_{n\to\infty} \sum\limits_{i=1}^{n} A_i = \frac{4}{3}(16)^2$

6. A circle S having centre (α, β) intersect a parabola $y^2 = 4x$ at three points A, B and C such that normals at these points are concurrent at (9, 6), where O is origin. The slope of normals at A & B are positive, then

(A) Sum of modulus of slopes of normals at points A, B and C is 6

(B) $|\alpha - \beta| = 4$

(C) point of intersection of tangents at A & B will be $(2, -3)$

(D) Circle S also passes through O

7. If $P = \sum_{r=3n}^{r=4n-1} \left[\frac{r^2 + 13n^2 - 7rn}{n^3}\right]$ & $Q = \sum_{r=3n+1}^{r=4n} \left[\frac{r^2 + 13n^2 - 7rn}{n^3}\right]$, then

(A) $P > \frac{5}{6}$

(B) $Q < \frac{5}{6}$

(C) $Q > \frac{5}{6}$

(D) $P < \frac{5}{6}$

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –1 In all other cases.

---

8. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Given all terms of an A.P. are positive integers, third term is 8 and sum of first 15 terms lies between 191 and 196, then first term is	(1)	1
(Q)	Suppose that for some angles x and y the equation $\cos 2y - \cos 2x = 3a - 2$ and $\cos^2 x + \sin^2 y = \frac{a^2}{2}$ hold simultaneously, then number of possible values of a is/are	(2)	4
(R)	Number of solution(s) of the equation $\sin^{-1}\frac{2x}{1+x^2} = \frac{\pi}{2} + \cos^{-1}x$	(3)	6
(S)	In ΔABC, minimum value of $\frac{1}{\Delta}\sqrt{abc(a+b+c)}$ is, (all symbols have their standard meaning)	(4)	0
		(5)	5

(A) P $\longrightarrow$ 1;Q $\longrightarrow$ 1;R $\longrightarrow$ 1;S $\longrightarrow$ 2

(B) P $\longrightarrow$ 3;Q $\longrightarrow$ 5;R $\longrightarrow$ 2;S $\longrightarrow$ 3

(C) P $\longrightarrow$ 3;Q $\longrightarrow$ 1;R $\longrightarrow$ 2;S $\longrightarrow$ 1

(D) P $\longrightarrow$ 3;Q $\longrightarrow$ 1;R $\longrightarrow$ 1;S $\longrightarrow$ 2

9. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Given $\int_{-2}^{2} f(\{x\})\,dx = 0$ and $\int_{-2}^{6} f(\{x\})\,dx = -2$ then value of $\int_{6}^{2} f(\{x\})\,dx$ is equal to (Here {.} denotes fractional part of function)	(1)	2
(Q)	If $\left\vert\vec{a}-\vec{b}\right\vert = 8$, $\vert\vec{b}\vert = \sqrt{57}$ and $\left\vert\vec{a}+\vec{b}\right\vert = 10$, then value of $\vert\vec{a}\vert$ is equal to	(2)	3
(R)	$\int_{-\pi}^{\pi} \frac{\vert\text{sgn}(\cot x) + \text{sgn}(\tan x)\vert\,dx}{\pi}$ is	(3)	4
(S)	Let $A = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & \frac{1}{2^2} \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & \frac{1}{2^3} \\ 0 & 1 \end{bmatrix} \ldots \infty$ times, then sum of all elements of matrix A is	(4)	5
		(5)	6

(A) P $\longrightarrow$ 5;Q $\longrightarrow$ 4;R $\longrightarrow$ 3;S $\longrightarrow$ 2

(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 4;R $\longrightarrow$ 3;S $\longrightarrow$ 2

(C) P $\longrightarrow$ 1;Q $\longrightarrow$ 4;R $\longrightarrow$ 2;S $\longrightarrow$ 2

(D) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 2

10. Suppose n people (n > 6) are asked a question successively in a random order and exactly 3 out of 'n' people know the answer. Let P_r denote the probability that r^{th} person asked, is the first to know the answer, then

List-I		List-II	
(P)	Probability that first four do not know the answer	(1)	$\frac{3(n-3)}{n(n-1)}$
(Q)	P_2	(2)	$\frac{(n-4)(n-5)(n-6)}{n(n-1)(n-2)}$
(R)	P_{n-2}	(3)	$\frac{6}{n(n-1)(n-2)}$
(S)	P_r	(4)	$\frac{3(n-r)(n-r-1)}{n(n-1)(n-2)}$
		(5)	$\frac{6(n-r)(n-r-1)}{n(n-1)(n-2)}$

(A) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 3;S $\longrightarrow$ 4

(C) P $\longrightarrow$ 2;Q $\longrightarrow$ 1;R $\longrightarrow$ 3;S $\longrightarrow$ 5

(D) P $\longrightarrow$ 1;Q $\longrightarrow$ 2;R $\longrightarrow$ 4;S $\longrightarrow$ 3

11.

List-I		List-II	
(P)	$\lim\limits_{x\to 0}\left[\left(\min\left(y^2+2y+4\right)\right)\dfrac{\sin x}{x}\right]$ is equal to, where [.] denotes G.I.F.	(1)	2
(Q)	$\lim\limits_{x\to 0}\dfrac{1}{x}.\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)=$	(2)	3
(R)	The number of points in the domain of the function $f(x)=\dfrac{1}{\ln\lvert x-4\rvert}$ is discontinuous are	(3)	6
(S)	Number of critical points of $f(x)=\min(\sin x,\cos x)$, in $x\in[0,2\pi)$, is	(4)	0
		(5)	5

(A) P $\longrightarrow$ 1;Q $\longrightarrow$ 1;R $\longrightarrow$ 2;S $\longrightarrow$ 5

(B) P $\longrightarrow$ 1;Q $\longrightarrow$ 1;R $\longrightarrow$ 4;S $\longrightarrow$ 5

(C) P $\longrightarrow$ 2;Q $\longrightarrow$ 4;R $\longrightarrow$ 5;S $\longrightarrow$ 2

(D) P $\longrightarrow$ 2;Q $\longrightarrow$ 4;R $\longrightarrow$ 1;S $\longrightarrow$ 3

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases.

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1. If 4-horses participate in race. The number of ways in which they can go through finish line is α (Dead heat is possible), then value of α is (Two or more horses can finish race exactly at same time)

2. A line $3x + y = 8$ touches a hyperbola $H = 0$ at P(1, 5) meets its asymptotes at A and B. If $AB = 2\sqrt{10}$, C(1, 1) be the centre of hyperbola, e and ℓ are eccentricity and latus rectum, (here length of conjugate axis < length of transverse axis), then value of $[e^2 + \ell^2]$ is (Where [.] Greatest Integer Function)

3. Lines $L_1 : \frac{x-6}{3} = \frac{y-4}{2} = \frac{z-2}{0}$ and $L_2 : \frac{x-8}{4} = \frac{y-2}{0} = \frac{z-4}{2}$ meets the plane $\pi : \vec{r}.(2\hat{i} + \hat{j} - \hat{k}) = 6$ at point A and B respectively. Area of the triangle formed by the lines L_1, L_2 and AB is

4. Let A and B are non-singular matrices of order 3 such that det. (A) = 5 and $A^{-1}B^2 + AB = O$. If value of $\det.(A^6 - 2A^4B + A^2B^2)$ is equal to 10^n, then value of n is

5. Let ω be the complex number representing the point M $\left(\frac{-1}{2}, \frac{\sqrt{3}}{2}\right)$, then number of distinct complex numbers z satisfying the equation $(z+1)\begin{vmatrix} z+\omega^2 & 1 \\ 1 & z+\omega \end{vmatrix} + \omega \begin{vmatrix} 1 & \omega \\ z+\omega & \omega^2 \end{vmatrix} + \omega^2 \begin{vmatrix} \omega & z+\omega^2 \\ \omega^2 & 1 \end{vmatrix} = 0$ is equal to :

6. Let f(x) be a twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(2-x)$. Then value of $\int_{-1}^{1} f'(1+x)x^2e^{x^2}\,dx$ is equal to

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



English

(0000CJA101021230017)

Test Pattern

ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
AIOT
21-04-2024

JEE(Main+Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

Time : 3 Hours

PAPER-2 (OPTIONAL)

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take $\mathbf{g = 10\ m/s^2}$ unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^{8}\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\epsilon_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

PART-1 : PHYSICS

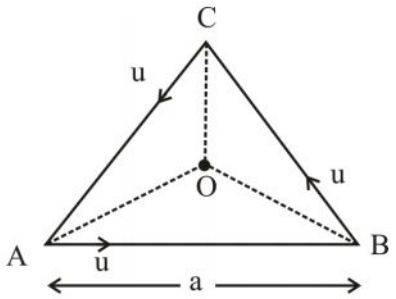
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. Three particles A, B and C situated at vertices of an equilateral triangle, all moving with same constant speed such that A always move towards B, B always towards C and C always towards A. Initial separation between each of the particle is a. O is the centroid of the triangle. Distance covered by particle A when it completes one revolution around O is



(A) $2a\left(1-e^{-2\sqrt{3}\pi}\right)$

(B) $\frac{2a}{3}\left(1-e^{-2\sqrt{3}\pi}\right)$

(C) $a\left(1+e^{-2\sqrt{3}\pi}\right)$

(D) $\frac{2a}{3}\left(1-e^{-\sqrt{3}\pi}\right)$

2. Use a diameter 1 mm superconductive wire to construct a circle of radius 5 cm. When the circular loop is superconductive, the current is 100 A. After one year, it is observed that the circular loop's change in current is less than 10^{-6} A. The resistivity of wire (in Ω-m) is of the order of :-

(A) 10^{-20}

(B) 10^{-25}

(C) 10^{-30}

(D) 10^{-35}

3. In cloudy weather, measurements are taken with a light meter pointed towards the sky. We find that, due to the scattering of light by the clouds, the incident power per unit area is approximately I_0, regardless of the orientation of the light meter. On top of an opaque thin walled spherical shell of radius R, there is a small hole of radius r (much larger than the wavelength of visible light). The inner side of the wall of the spherical shell is coated with soot. The intensity of illumination on the inner surface of the shell, when it is placed in the open air is :-

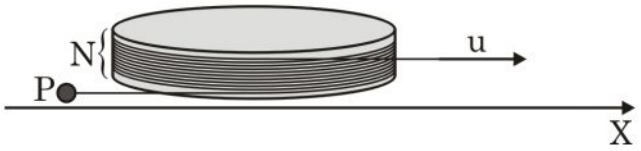
(A) $I_0\frac{r^2}{R^2}$

(B) $I_0\frac{r^2}{4R^2}$

(C) $I_0\frac{r^2}{R^2-r^2}$

(D) $I_0\frac{r^2}{R^2+r^2}$

4. A massless thread makes N turns around statically fixed cylinder, as shown in the figure. Initially, the free (unwound) ends of the thread are parallel to the axis X. Then, a heavy point-like object P is attached to one end of the thread while the other end is pulled with a constant velocity u along X. Find the velocity attained by the heavy object when it detaches from the cylinder. Suppose that the turns of the thread are wound tightly to one another and are placed practically in the same plane, perpendicular to the cylinder axis. Neglect any friction in the system. Do not consider the force of gravity.



(A) u

(B) $2\pi Nu$

(C) $\sqrt{1+(2\pi N)^2}u$

(D) πNu

SECTION-I (ii) : (Maximum Marks: 12)

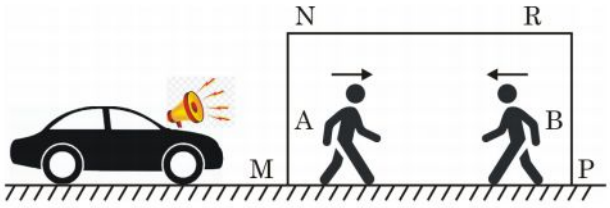
- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. A car is moving with a velocity $\frac{3v}{10}$ towards a medium MNRP as shown, where 'v' is the speed of sound in air. Two men A and B are moving towards each other in the medium with the velocities $\frac{v}{6}$ and $\frac{v}{9}$ respectively. The speed of sound in medium is 7v and medium is kept at rest. The car is emitting a sound of frequency f. Then choose the correct option(s).



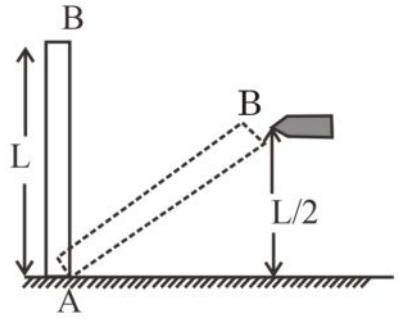
(A) The wavelength received by the man A is $\frac{49v}{10f}$

(B) The wavelength received by the man A is different from the wavelength received by man B

(C) The ratio of frequency observed by the man A to that observed by the man B is $\frac{123}{128}$

(D) The frequency observed by the man B is $\left(\frac{640}{441}\right)$ f

6. A rod of mass 'M' and length 'L' was at rest in a vertical position, it's end A being hinged about which it can freely rotate. A slight disturbance caused the rod to start rotating clockwise. A bullet of mass 'm' and speed 'v_0' flying horizontally at a height 'L/2' above ground hits the end B of the rod and gets embedded. (Take M = 3m). Then choose the correct statement(s).



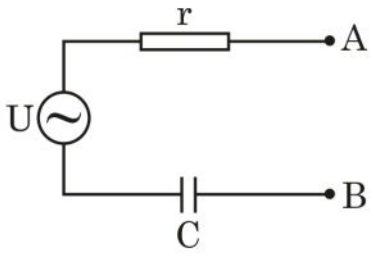
(A) Minimum velocity of the bullet such that the rod again becomes vertical is $v_0 = \sqrt{gL}\left(\sqrt{6}+\sqrt{20}\right)$

(B) Tangential acceleration of the midpoint of the rod when it makes an angle 60° with horizontal after collision is $\frac{5g}{16}$

(C) Minimum velocity of the bullet such that the rod again becomes vertical is $v_0 = \sqrt{3gL}\left(\sqrt{6}+\sqrt{20}\right)$

(D) Tangential acceleration of the midpoint of the rod when it makes an angle 60° with vertical after collision is $\frac{5g}{12}$

7. The equivalent circuit of a real source of alternating voltage of frequency $\omega = 1.00 \times 10^3\ s^{-1}$ consists of an ideal voltage source with the amplitude U = 15.0 V, a resistor with the resistance r = 10Ω and a capacitor with the capacitance C = 100 μF. Different circuits of resistors, capacitors and coils can be connected to the source as a load. A load circuit, which assures maximum heat generated in the load itself is connected across the source terminals AB.



(A) The load is purely resistive in nature.

(B) The power factor of the load is $\frac{1}{\sqrt{2}}$.

(C) The resistance of the load is equal to 10Ω.

(D) The maximum heat generated in the load is $\frac{45}{32}$W.

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

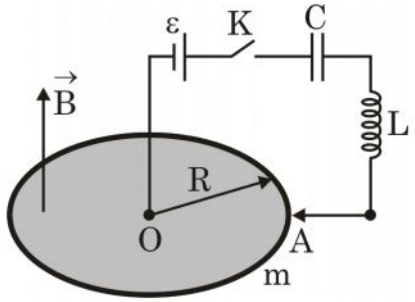
A solid cylinder of mass m and radius r rests on the inside of a thin-walled cylinder of mass M and radius R. The solid cylinder is made to oscillate around the equilibrium position.
(Take M = 1kg, m = 1 kg, R = 1m and r = 0.25 m)

1. The angular frequency (in rad/s) of oscillation if the large cylinder is fixed and small cylinder rolls without slipping in the bottom of the cylinder is :-

2. The angular frequency (in rad/s) of oscillation if the large cylinder is free to rotate about an axis through its fixed centre and small cylinder rolls without slipping in the bottom of the cylinder.

Paragraph for Questions 3 and 4

The figure shows a diagram of an electrical circuit consisting of a source with electromotive force ε, a capacitor with capacitance C, a coil of inductance L, a key K and a metal disk of radius R. The disk can rotate around its axis without friction. The disk is located in a uniform magnetic field with induction B, directed along its axis as shown in the figure. Almost the entire mass of the disk is concentrated in its thin rim. The source is connected to the disk at its centre O, and the inductor is connected to the disk using a fixed sliding contact A. Initially, key K is open, the disk is motionless, there is no current in the coil, and capacitor is not charged. Key K is closed.
(Take : B = 1T, C = 1F, ε = 1V, m = 1 kg, R = 1m, L = 1H, π = 3.14 and $\sqrt{5}$ = 2.24)



3. The maximum angular velocity (in rad/sec) of the disc after key is closed is :-

4. The time (in sec) after closing the key after which the angular velocity of the disk first reaches the maximum value is :-

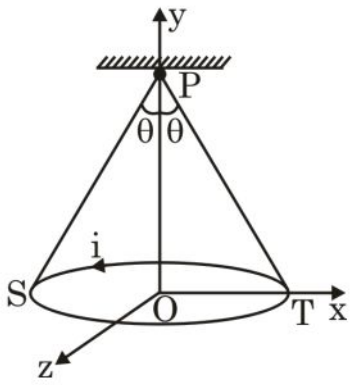
SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

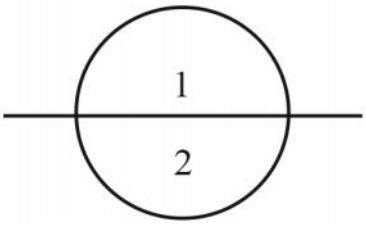
Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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5. A ring of uniform mass density of mass m (m = 100 gm) and radius R (R = 50 cm) is suspended by two insulated strings PS and PT from the point P on y-axis in vertical plane. The ring is kept parallel to horizontal surface on x-z plane. Ring carries a current $\left(i = \frac{7}{22}\text{amp}\right)$ as shown in the figure. There exist a magnetic field (B = 1 Tesla) along negative x-axis. The tension in the string PS is T_1 and in the string PT is T_2. Find the ratio $\frac{T_1}{T_2}$. (It is given that $\theta = 60°$)



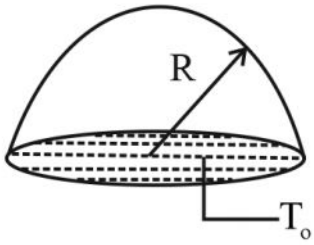
6. If we consider earth as a uniform solid sphere of mass M and Radius R then force of interaction between two hemi-spheres will be $\frac{3GM^2}{\alpha R^2}$. Find α



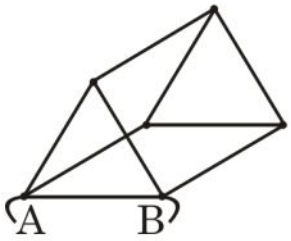
7. A given sample contains two types of atoms A and B in the ratio 3 : 1. Atoms of type A undergo α-decay with a half life of 30 days to form 'B' while atoms of type B undergo α-decay with a half life of 45 days to form 'C', which is stable. Calculate the time (in days) after which the activities of A and that of B are in the ratio 9 : 22. If time is T write $\frac{T}{10}$.

$$A \xrightarrow{T_{1/2} = 30 \text{ days}} B \xrightarrow{T_{1/2} = 45 \text{ days}} C$$

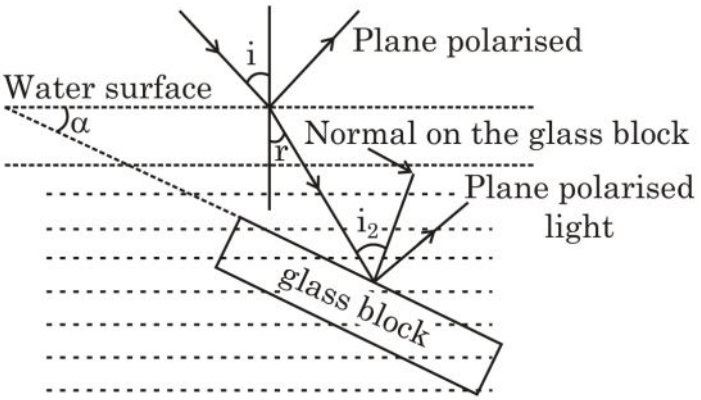
8. Circular base of hemi-spherical shell base closed body is maintained at $T_o = 30K$, outer atmosphere is at 0 K. If equilibrium temperature of curved surface is $10 \times (x)^{\frac{3}{4}}K$, write x :-



9. The resistance between two joints is R. The equivalent resistance between A and B is $\frac{\alpha R}{15}$. The value of α is :-



10. Unpolarised light is incident on a water surface at the Brewster angle. The refracted ray meets the block of glass $\left[\mu_g = \frac{3}{2}\right]$ immersed in the water $\left(\mu_w = \frac{4}{3}\right)$. The flat upper surface of the block makes an angle α with water surface. The value of α for which the light reflected from the block is completely polarized. $(\tan^{-1}\frac{9}{8} = 48.4)$ Fill the value of 10α.



PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

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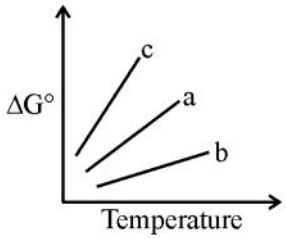
1. The standard Gibb's free energy of the following reaction, ΔG° have been determined at various temperatures :

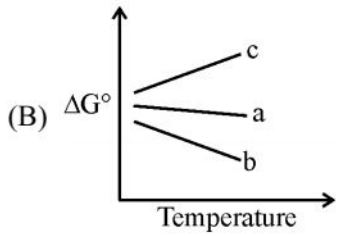
(a) $C(s) + O_2(g) \longrightarrow CO_2(g)$

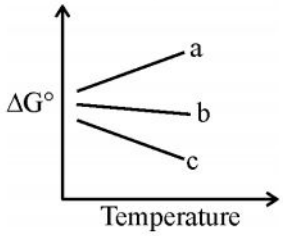
(b) $C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g)$

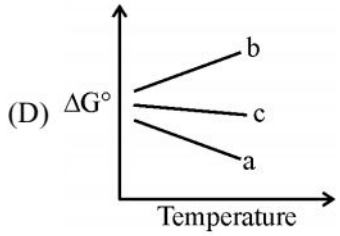
(c) $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$

Which of the following plots would represent most likely the temperature dependence of ΔG° ?

(A) 

(B) 

(C) 

(D) 

2. Potential energy of electron present in 2nd orbit of Li^{2+} is : (a_0 = Radius of 1st Bohr's orbit) :

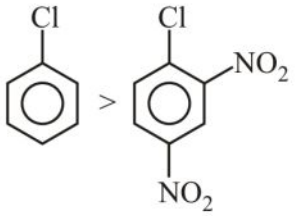
(A) $-\frac{e^2}{4\pi\varepsilon_0 a_0}$

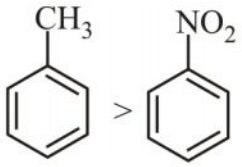
(B) $-\frac{3e^2}{4\pi\varepsilon_0 a_0}$

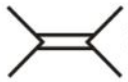
(C) $-\frac{3e^2}{16\pi\varepsilon_0 a_0}$

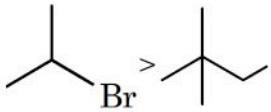
(D) $-\frac{9e^2}{16\pi\varepsilon_0 a_0}$

3. Which of the following is not correct order of reactivity.

(A)  (towards NaOH)

(B)  (towards EAS reaction)

(C)  $> CH_2 = CH_2$ (towards HBr)

(D)  (towards NaI, acetone)

4. Consider the following reaction/s & choose **INCORRECT** statement.

$2NaCl + 2CO_2 + 2NH_3 + 2H_2O \longrightarrow (P) + (Q)$

$(P) \xrightarrow{\Delta} (R) + CO_2 + H_2O$

(A) In solid salt (P) anion form an infinite chain

(B) From compound (Q) NH_3 can be obtained by adding $Ca(OH)_2$

(C) Compound (P) has symmetrical H-bond

(D) MnO_2 salt give green coloured bead when heated with compound (R) in presence of air.

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. At 298 K, standard reduction potential is 1.33 V for $Cr_2O_7^{2-}/Cr^{3+}$.

Select the correct statement(s). (If concentration of other species are 1M)

Given : $\frac{2.303R \times 298}{F} = 0.06$ V, $E^{o}_{Br_2/Br^-} = 1.07$ V and $E^{o}_{I_2/I^-} = 0.54$ V.

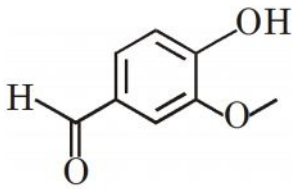
(A) $Cr_2O_7^{2-}$ may oxidise Br^- and I^- at pH = 3

(B) $Cr_2O_7^{2-}$ may oxidise Br^- and I^- at pH = 0

(C) $Cr_2O_7^{2-}$ may oxidise I^- but can't oxidise Br^- at pH = 3

(D) $Cr_2O_7^{2-}$ may oxidise I^- but can't oxidise Br^- at pH = 0

6. Choose the **CORRECT** statement(s) regarding Vanilin



Vanilin [P]

(A) **[P]** can be obtained as a major product when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH and one equivalent of MeBr.

(B) **[P]** can be obtained as a one of the product when 2-methoxy phenol is treated with chloroform and aq KOH.

(C) Positional isomer of **[P]** can be obtained when 3,4-dihydroxy benzaldehyde is treated with one equivalent of NaOH followed by treatment with one equivalent of MeBr.

(D) **[P]** can be obtained when 3,4-dihydroxy benzaldehyde is treated with two equivalent of NaOH and one equivalent of MeBr followed by protonation.

7. Choose correct statement/s about extraction of various metal in metallurgical process.

(A) In Mond process the decomposition of nickel carbonyl is accompanied by a large entropy increase (four moles of gas formed from one mole of complex) and hence it takes place spontaneously at higher temperature when $T\Delta S$ term becomes significant

(B) Crude gold obtained by Mac Arthur Forest cyanide process contain Zn and sometime Pb impurity

(C) In Van Arkel-De Boer process the impure element is heated with iodine, producing a volatile iodide.

(D) In Kroll process

$$TiCl_4 + 2Mg \xrightarrow{\text{under atomosphere of Ar}} Ti + 2MgCl_2$$

Here $MgCl_2$ is leached with H_2O

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

For the reaction, $CHCl_3(g) + Cl_2(g) \xrightarrow{h\nu} CCl_4(g) + HCl(g)$ mechanism is given below

$Cl_2(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2Cl(g)$

$CHCl_3(g) + Cl(g) \xrightarrow{k_2} CCl_3(g) + HCl(g)$

$CCl_3(g) + HCl(g) \xrightarrow{k_3} CCl_4(g)$

Given :

$k_1 = 4.8 \times 10^3\ sec^{-1}$, $k_{-1} = 1.2 \times 10^3\ M^{-1}\ sec^{-1}$,

$k_2 = 1.3 \times 10^{-2}\ M^{-1}\ sec^{-1}$, $k_3 = 2.1 \times 10^2\ M^{-1}\ sec^{-1}$

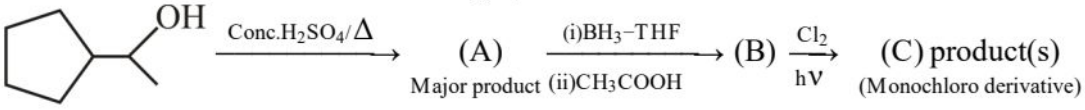
$E_{a_1} = 10$ kJ/mol, $E_{a_{-1}} = 15$ kJ/mol,

$E_{a_2} = 45$ kJ/mol, $E_{a_3} = 12$ kJ/mol

1. Overall rate constant of reaction, $CHCl_3(g) + Cl_2(g) \xrightarrow{h\nu} CCl_4(g) + HCl(g)$, is _____ $\times 10^{-2}\ M^{-1/2}\ sec^{-1}$.

2. Overall activation energy of reaction, $CHCl_3(g) + Cl_2(g) \xrightarrow{h\nu} CCl_4(g) + HCl(g)$, is _____ kJ/mole.

Paragraph for Questions 3 and 4



3. Total number of fractions obtained on fractional distillation of all monochloro derivatives product(s) (C)?

4. Calculate mass of Cl_2 required to react with 9.8 gm of compound (B) to form monochloro product. Assuming % yield = 100%

SECTION-II (ii) : (Maximum Marks: 24)

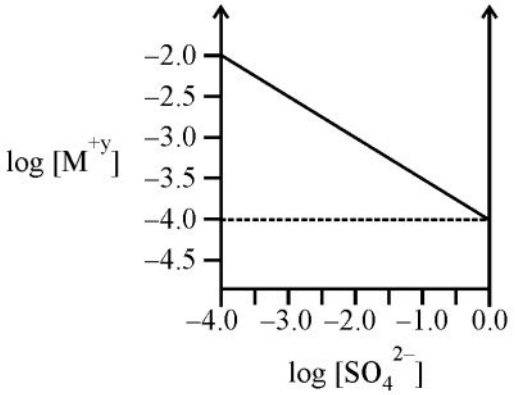
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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5. A metal M form sparingly soluble metal sulphate. The logarithm of concentration of dissolved metal ion in a solution saturated in the metal sulphate is plotted below as a function of logarithm of concentration of sulphate ion in solution. The molecular formula of metal sulphate is $M_x(SO_4)_y$. Then find $z \times 10^{-6}$.

$z = \dfrac{x+y}{K_{sp} \text{ of } M_x(SO_4)_y}$, x and y are minimum integer co-efficient.



6. Total number of Pseudohalides ion which are planar and polar.

$\overset{\Theta}{C}N$, $S\overset{\Theta}{C}N$, $Se\overset{\Theta}{C}N$, $O\overset{\Theta}{C}N$, CN_2^{2-} , $O\overset{\Theta}{N}C$, N_3^{Θ} , I_3^{Θ} , ICl_4^{Θ}

7. Total number of reaction/s in which paramagnetic gaseous product form on heating.

(1) $KMnO_4 \xrightarrow{\Delta}$

(2) MnO_2 + conc. HCl $\xrightarrow{\Delta}$

(3) MnO_2 + conc. $H_2SO_4 \xrightarrow{\Delta}$

(4) $KClO_3 \xrightarrow{\Delta}$

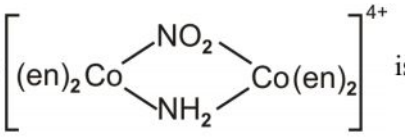
(5) $Pb(NO_3)_2 \xrightarrow{\Delta}$

(6) $CaCO_3 \xrightarrow{\Delta}$

(7) $NaNO_3 \xrightarrow{\Delta}$

(8) $NH_4NO_3 \xrightarrow{\Delta}$

(9) Na_2CrO_4 + conc. $H_2SO_4 \longrightarrow$

8. Total number of stereoisomer possible for complex compound  is.

9. $NaCl_{(s)} + K_2Cr_2O_{7(s)} + \text{conc. } H_2SO_4 \xrightarrow{\text{Heat}}$ Reddish brown fumes

$\xrightarrow{NaOH_{(aq.)}}$ Yellow solution (A)

Total number of reagent/s which give redox reaction with solution (A) in acidic medium among the following.

$Pb(CH_3COO)_2$, $Ba(NO_3)_2$, NO_2^- , Sn^{2+} , SO_2 , Fe^{2+} , H_2S , I^- , C_2H_5OH

10. How many amino acids are positively charged at pH = 7.

(i) Asparagine (ii) Valine (iii) Aspartic acid

(iv) Glycine (v) Lysine (vi) Glutamine

(vii) Glutamic acid (viii) Arginine (ix) Methionine

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered)
Negative Marks	:	–1	In all other cases

---

1. Let $\lim\limits_{n\to\infty} \frac{1}{n}\left\{\left(\sum\limits_{r=1}^{n} \ell n\left(n^2+r^2\right)\right) - 2n\ \ell n\ n\right\} = \ell n2 + \frac{\pi}{2} - 2$

If $\lim\limits_{n\to\infty} \frac{1}{n^{2m}}\left[\left(n^2+1^2\right)^m\left(n^2+2^2\right)^m\left(n^2+3^2\right)^m \ldots \left(n^2+n^2\right)^m\right]^{\frac{1}{n}} = \left(\frac{a\sqrt{e^{\pi}}}{e^2}\right)^m$, then the value of a is (where m, n $\in$ N)

(A) 1

(B) $\frac{3}{2}$

(C) 2

(D) $\frac{1}{2}$

2. The remainder when $\sum\limits_{k=0}^{1012}\left[\frac{{}^{1012}C_k}{{}^{2024}C_k}\sum\limits_{r=k}^{2024} {}^{r}C_k \cdot {}^{2024}C_r\right]$ is divided by 25 is (where ${}^{n}C_r = \frac{n!}{r!\,(n-r)!}$)

(A) 15

(B) 17

(C) 21

(D) 11

3. A moving particle starts at the point (4, 4) and moves until it hits one of the coordinate axes for the first time. When the particle is at the point (a, b) it moves at random to one of the points (a − 1, b), (a, b − 1) or (a − 1, b − 1) each with probability $\frac{1}{3}$, independent of its previous moves. The probability that it will hit the coordinate axes at (0, 0) is $\frac{x}{3^y}$. Where x and y are positive integers such that x is not divisible by 3. Then value of x + y is

(A) 252

(B) 253

(C) 152

(D) 353

4. Let H be the hyperbola having eccentricity 'e' and C be the smallest circle touching H and its pair of asymptotes. Ratio of radius of C with the length of semi-conjugate axis of H is.

(A) $e + \sqrt{e^2 - 1}$

(B) $e - \sqrt{e^2 - 1}$

(C) $e - \sqrt{e^2 + 1}$

(D) $\frac{1}{e}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	–2	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

---

5. Let A and B be two 3×3 matrices where both A and $I + AB$ are non-singular. If $3A^2B^2 + AB = 2(AB)^2$ then, which of the following is greater than $|B|$ (where I is identity matrix of order 3 & $|X|$ denotes determinant of X)

(A) 1

(B) 2

(C) 3

(D) 4

6. Consider $f : \left[0, \frac{\pi}{2}\right] - \left\{\frac{\pi}{4}\right\} \to R$. Let $f(x) = \lim\limits_{n\to\infty} \prod\limits_{k=0}^{n} \frac{\left((\cot x)^{2^k} + 1\right)^2}{\left((\cot x)^{2^{k+1}} + 1\right)}$, $\forall x \in \left(0, \frac{\pi}{2}\right] - \left\{\frac{\pi}{4}\right\}$ and $f\left(\frac{\pi}{4}\right) = 2$ and $f(0) = 0$, Then which of the following is true?

(A) f(x) is increasing in $\left(0, \frac{\pi}{4}\right)$ & $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(B) f(x) is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(C) f(x) has one local maxima

(D) f(x) has three local minima

7. If $z = e^{i\pi/13}$ then $\frac{1}{1-z}$ is equal to (Where $i = \sqrt{-1}$)

(A) $1 - z^2 + z^4 - z^6 + z^8 - z^{10} + z^{12}$

(B) $1 + z^2 + z^4 + z^6 + z^8 + z^{10} + z^{12}$

(C) $z - z^3 + z^5 - z^7 + z^9 - z^{11}$

(D) $z + z^3 + z^5 + z^7 + z^9 + z^{11}$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **TWO (02)** paragraphs.
- Based on each paragraph, There are **TWO (02)** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider a circle C : $x^2 + y^2 = 5$,

Plane π_1 : $x - y + 2z + 5 = 0$ and

Plane π_2 : $2x + y + z - 5 = 0$ and

Plane π_3 : $x + 2y - z = 0$.

Let reflection of curve 'C' about plane mirror π_1 be curve C_1. [.] represents G.I,F.

1. If λ is the shortest distance of plane π_2 from any point on C_1 then $[\lambda]$ equal to:

2. Planes π_1, π_2, π_3, form a prism. If 'A' is area of cross-section of this prism then [A] equal to

Paragraph for Questions 3 and 4

Consider $\int_0^{\frac{\pi}{2}} (\sin x)^m (\cos x)^n dx, m, n \in N$.

We can simplify this integration by substituting sinx = t OR cosx = t.

Based on this solve below questions

3. $I_1 = \int_0^1 x^6(1-x)^4 dx, I_2 = \int_0^1 x^5(1-x)^3 dx$

then $\frac{I_1}{I_2} = \frac{p}{q}$, where p, q are co-prime positive integers then p + q equal to

4. Let $I = \int_0^{\frac{\pi}{4}} \frac{\left(\cos\left(x + \frac{\pi}{4}\right)\right)^{10} . (\sin x)^7}{(\cos x)^{19}} dx = \frac{m!n!}{2^p.18!}$, m, n, p are natural numbers then smallest value of

m + n + p = ?

SECTION-II (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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5. Let us consider a complex number $z = x^{3/2} + iy^2; x \geqslant 0; y \in R$ such that $2x^5y - x^4y + 2x^3y - x^2y + 4xy - 2y = 0$ and $|z|^2 = (\text{Re}(z))^{2/3}$. Then number of all distinct value(s) of $|z|$ is

6. Consider a circle $S : x^2 + y^2 + x - y = 2021$. If (x, y) is called integral points if x and y are both integer. Then number of integral points on $S = 0$ is

7. Find number of permutations x_1, x_2, x_3, x_4, x_5 of numbers 1, 2, 3, 4, 5 taken all at a time such that the sum of five products $x_1x_2x_3 + x_2x_3x_4 + x_3x_4x_5 + x_4x_5x_1 + x_5x_1x_2$ is divisible by 3.

8. The third derivative of $\frac{x^4}{(x-1)(x-2)}$ is $-\frac{k}{(x-2)^4} + \frac{6}{(x-1)^4}$. The number of $\theta \in [0, 2\pi]$ satisfying the equation $\sin^2\theta + 2\cos^2\theta = k - 96 - 3\sin\theta\cos\theta$ is ____.

9. If $A = \begin{bmatrix} 6 & 3 \\ 2 & 1 \end{bmatrix}$ and $10^3 < \text{tr}(A^n) < 10^4$ where $n \in N$. The value of n is (where tr(X) denotes trace of matrix X)

10. Let us consider the curves $9x^2 + 4y^2 = 900$, $x^2 + y^2 - 4x - 6y - 12 = 0$ and $y^2 - 6y - 6x + 51 = 0$. The maximum number of disjoint regions into which these curve divide xy plane (excluding themselves).

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work



ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Advanced)
AIOT
21-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	D	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	B,C	A,B,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	C	A	C		
SECTION-II	Q.	1	2	3	4	5	6
	A.	594	4	3	1	1	22

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	B	D	D		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	B,C	A,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	C	A	B	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	4	160	3	331	3	2

PART-3 : MATHEMATICS

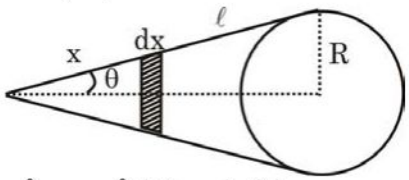
SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	C	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C	A,B,C,D	A,C			
SECTION-I (iii)	Q.	8	9	10	11		
	A.	D	B	A	B		
SECTION-II	Q.	1	2	3	4	5	6
	A.	75	3	1	6	1	0

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)



$$\int dv = \int \frac{k\,2\pi x \sin\theta dx\sigma}{x}$$

$$V = 2\pi\sigma k \sin\theta \ell = 2\pi\sigma k \frac{R}{\ell}\ell$$

$$= 2\pi\sigma R \times \frac{2}{4\pi\varepsilon_0} = \frac{\sigma R}{2\varepsilon_0}$$

$$V(\text{after removal of part AB}) = \frac{5}{6} \times \frac{\sigma R}{2\varepsilon_0}$$

$$= \frac{5\sigma R}{12\varepsilon_0} = \frac{5 \times R \times 12\varepsilon_0}{12\varepsilon_0} = 5V$$

2. Ans (C)

$$J \cdot \frac{a}{\sqrt{2}} = \frac{(2m)a^2}{12} \cdot \omega$$

$$\Rightarrow \omega = \frac{3\sqrt{2}}{ma} J$$

$$2T = m \cdot \omega^2 \cdot \frac{a}{3\sqrt{2}} \Rightarrow T = \frac{3J^2}{\sqrt{2}ma}$$

3. Ans (D)

Gain in PE of ejected part = $(\rho a^3)g\left(\frac{a}{2}\right) \propto a^4$

KE of meteor $\propto a^4$

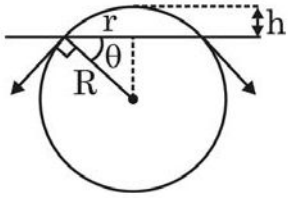
$a \propto (KE)^{1/4}$

Volume $\propto a^3 \propto (KE)^{3/4}$

as mass doubles, KE doubles and volume becomes

$2^{3/4}$ times.

4. Ans (C)



$$\left(\frac{V}{d}\right) \cdot \left(\frac{V \epsilon_0}{d} \pi r^2\right) \times \frac{1}{2} = \sigma \times 4\pi r \times \cos\theta$$

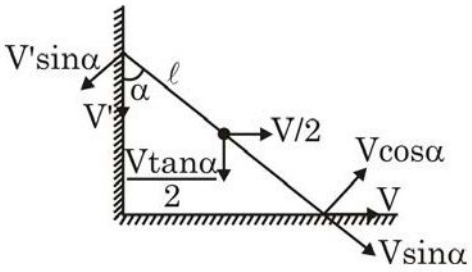
$$\Rightarrow \cos\theta = \frac{\epsilon_0 V^2 r}{8\sigma d^2}$$

$$h \approx \frac{1}{2} r \cos\theta = \frac{\epsilon_0 V^2 r^2}{16\sigma d^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C)



$V \sin\alpha = V' \cos\alpha$

$V' = V \tan\alpha$

$$\frac{d\alpha}{dt} = \frac{V \cos\alpha + V' \sin\alpha}{2\ell}$$

$$\frac{d\alpha}{dt} = \frac{V \cos\alpha + V \tan\alpha \sin\alpha}{2\ell}$$

$$\frac{d\alpha}{dt} = \frac{V}{2\ell \cos\alpha}$$

Horizontal acceleration of sphere = 0

$$\text{Vertical acceleration} = d\frac{\left(V\frac{\tan\alpha}{2}\right)}{(dt)} = \frac{V}{2}\sec^2\alpha\frac{d\alpha}{dt}$$

$$= \frac{V}{2}\sec^2\alpha \times \frac{V}{2\ell\cos\alpha}$$

$$\text{Vertical acceleration} = \frac{V^2}{4\ell} \times \frac{1}{\cos^3\alpha}$$

$$mg - N = \frac{mV^2}{4\ell\cos^3\alpha}$$

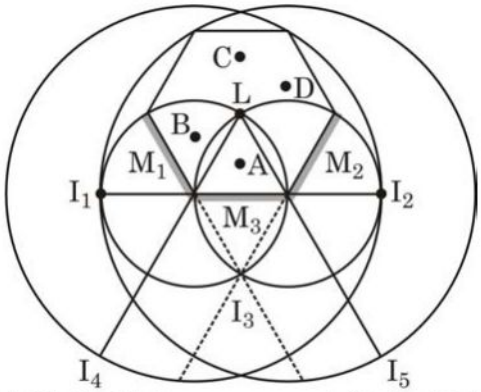
$$N = mg - \frac{mV^2}{4\ell\cos^3\alpha}$$

$$= mg - \frac{mV^2 2\sqrt{2}}{4\ell} = mg - \frac{mV^2}{\sqrt{2}\ell}$$

$$= m\left[g - \frac{V^2}{\sqrt{2}\ell}\right]$$

$$\text{Vertical acceleration } (\alpha = 60) = \frac{V^2}{4\ell}8 = \frac{2V^2}{\ell}$$

6. Ans (B,C)



(of I_2 by M_1) (of I_1 by M_2)

A lies in the field of view of I_1, I_2 and I_3, but not of I_4 & I_5.

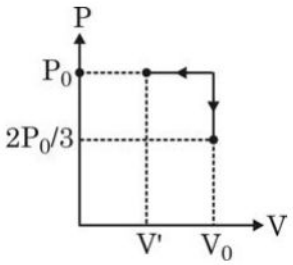
B lies in the field of view of I_1, I_2, I_3 and I_4, but not of I_5.

C lies in the field of view of all $-I_1$, I_2, I_3, I_4 and I_5.

D lies in the field of view of I_1, I_2, I_3 and I_5 but not of I_4.

7. Ans (A,B,C)

In 1st case, as the air enters slowly, there will be no change in its temperature (T_0). The air will fill until the pressure become $\frac{2P_0}{3}$. After releasing the piston (massless), due to outside and inside pressure difference, the piston suddenly falls down and the gas is compressed suddenly. The P–V graph of the process can be given as



$\Delta Q = 0$

$$\Rightarrow -P_0 \cdot (V_0 - V') + n \cdot \frac{5R}{2}(T - T_0) = 0$$

$$\Rightarrow -P_0V_0 + P_0V' + \frac{5}{2}nR(T - T_0) = 0$$

$$\Rightarrow -\frac{3}{2}nRT_0 + nRT + \frac{5}{2}nRT - \frac{5}{2}nRT_0 = 0$$

$$\Rightarrow \frac{7}{2}T = \frac{8}{2}T_0 \Rightarrow T = \frac{8}{7}T_0$$

In 2nd case, as the air enters through valve, the work done by atmospheric air on the entering air goes in increasing internal energy of entering air.

$$\Rightarrow P_0 \cdot \left(\frac{nRT_0}{P_0}\right) = n \cdot \frac{5R}{2} \cdot (T - T_0)$$

$$T_0 = \frac{5}{2}T - \frac{5}{2}T_0 \Rightarrow T = \frac{7}{5}T_0$$

Now, when the piston is released, final temperature will become $\frac{8}{5}T_0$.

PART-1 : PHYSICS

SECTION-I (iii)

8. Ans (D)

(P) : $\Delta x_1 = CP_1\frac{d}{D} = 0.3\mu m = \frac{3}{4}\lambda$

$\Delta x_2 = CP_2\frac{d}{D} = 1.2\mu m = 3\lambda$

$\Delta x_C = 0$

$\Rightarrow I_C = 4I_0,\ I_{P_1} = 2I_0,\ I_{P_2} = 4I_0$

(Q) : $\Delta x_C = 0 \Rightarrow I_C = 4I_0$

$\Delta x_1 = \mu CP_1\frac{d}{D} = 0.4\mu m = \lambda$

$I_{P_1} = 4I_0$

$\Delta x_2 = \mu CP_2\frac{d}{D} = 1.6\mu m = 4\lambda$

$I_{P_2} = 4I_0$

(R) : $\Delta x_C = (\mu - 1)t = 0.4\ \mu m = \lambda$

$I_C = 4I_0$

$\Delta x_1 = 0.3\ \mu m - 0.4\ \mu m = -0.1\ \mu m = \frac{-\lambda}{4}$

$\Rightarrow I_{P_1} = 2I_0$

$\Delta x_2 = 0.4\ \mu m - 0.4\ \mu m = 0$

$\Rightarrow I_{P_2} = 4I_0$

(S) : $\Delta x_C = d\sin\theta = d\theta = 0.1\ \mu m = \frac{\lambda}{4}$

$\Rightarrow I_C = 2I_0$

$\Delta x_1 = 0.3\ \mu m - 0.1\ \mu m = 0.2\ \mu m = \frac{\lambda}{2}$

$\Rightarrow I_{P_1} = 0$

$\Delta x_2 = 0.4\ \mu m - 0.1\ \mu m = 0.3\ \mu m = \frac{3\lambda}{4}$

$\Rightarrow I_{P_2} = 3I_0$

9. Ans (C)

$mvr = \frac{nh}{2\pi}$

$\frac{mv^2}{r} = \frac{K}{r^2}$

$mv^2r = K$

$v \propto \frac{1}{n}$

$r \propto n^2$

$\omega \propto \frac{1}{n^3}$

$T \propto n^3$

10. Ans (A)

$\epsilon_1 = \frac{x}{\ell}$

$\epsilon_2 = \frac{x}{4\ell}$

$\frac{\epsilon_1}{\epsilon_2} = 4$

$\sigma_1 = \sigma_2$

$\Rightarrow y_1\epsilon_1 = y_2\epsilon_2 \ \Rightarrow\ 4y_1 = y_2$

$u = \frac{1}{2y}\left(\frac{T}{A}\right)^2$

$U = \frac{T^2}{2Ay}.\ell$

$U_1 = U_2$

$\Rightarrow \frac{1}{2}\epsilon_1^2\, y_1 \times A\ell = \frac{1}{2}\epsilon_2^2\, y_2 \times A \times 2\ell$

$\frac{y_2}{y_1} = \frac{1}{2}\left(\frac{\epsilon_1}{\epsilon_2}\right)^2 = 8$

(S) $u_1 = u_2$

$\frac{1}{2}\times \epsilon_1^2 \times y_1 = \frac{1}{2}\times \epsilon_2^2 \times y_2$

$\frac{y_2}{y_1} = \left(\frac{\epsilon_1}{\epsilon_2}\right)^2 = 16$

11. Ans (C)

In KL, V = constant, W = 0, Q > 0

In LM, T = constant, W > 0, Q > 0

In ML, P = constant, W < 0, Q < 0

In JK, T = constant, W < 0, Q < 0

PART-1 : PHYSICS

SECTION-II

1. Ans (594)

$$\text{Least count} = \frac{P}{N} = \frac{1\text{mm}}{100} = 0.01\text{ mm}$$

Diameter of the wire is

$d = 4\text{mm} + 56 \times \text{L.C} - 6 \times \text{L.C}$

$d = 4\text{mm} + 50 \times \text{L.C}$

$d = 4\text{mm} + 50 \times 0.01\text{ mm}$

$d = 4.50\text{mm}$

$d = 0.450\text{ cm}$

The curved surface area of the wire is

$S = \pi dl$

$S = \frac{22}{7} \times 0.450 \times 4.2 = 5.94\text{ cm}^2$

2. Ans (4)

The tower will be stable if its center of mass is located within its area of support. If we set the origin of our coordinate system to be the bottom left corner, that's equivalent to saying

$x_{cm} \leqslant b$

Calculating x_{cm} :

$$x_{cm} = \frac{\frac{b}{2} + \frac{b}{2} + \frac{a}{2}}{2} = \frac{2b + a}{6}$$

Yields :

$2b + a \leq 6b$

$\frac{a}{b} \leqslant 4$

3. Ans (3)

$$\sqrt{2g(1-x)h} \times \sqrt{\frac{2xh}{g}} + \sqrt{2g(1-x)h} \times \sqrt{\frac{2(1+x)h}{g}}$$

$$= 2h\sqrt{x(1-x)} + 2h\sqrt{1-x^2}$$

$$y = \sqrt{x(1-x)} + \sqrt{1-x^2}$$

$$\frac{dy}{dx} = 0 = \frac{1-2x}{2\sqrt{x(1-x)}} + \frac{-2x}{2\sqrt{1-x^2}}$$

$$\frac{1-2x}{\sqrt{x(1-x)}} = \frac{2x}{\sqrt{1-x^2}}$$

$$\frac{(1-2x)^2}{x} = \frac{4x^2}{1+x}$$

$$4x^3 = (1+x)(1+4x^2-4x)$$

$$0 = -3x + 1 \Rightarrow x = \frac{1}{3}$$

4. Ans (1)

The magnetic field from the wire is given by

$$B = \frac{\mu_0 i_1}{2\pi x}$$

Let θ be the direction of a component of force from the vertical. It is then seen that

$dF = Bi_2 d\ell \Rightarrow dF_x = Bi_2 d\ell \sin\theta = Bi_2 dy$

We only consider the force in the x-direction which means that

$$F_x = \int_{\infty}^{\infty} dF_x = \frac{\mu_0 i_1 i_2}{2\pi} \int_{-\infty}^{\infty} \frac{dy}{x}$$

Solving the equation in terms of x and then plugging in gives us

$$F_x = \frac{8\mu_0 i_1 i_2}{2\pi} \int_{-\infty}^{\infty} \frac{dy}{y^2 - 6y + 25}$$

$$= \frac{8\mu_0 i_1 i_2}{2\pi} \cdot \frac{\pi}{4} = \mu_0 i_1 i_2$$

5. Ans (1)

$$m_1m_2 = \frac{v_1}{-u} \times \frac{-(u+d)}{(v_1-d)} = -1$$

$$1 + \frac{d}{u} = \frac{d}{v_1} - 1$$

$$2 + \frac{d}{u} = \frac{d}{v_1}$$

$$v_1 = \frac{ud}{d+2u} \quad \text{... (1)}$$

$$v_1 - d = \frac{-ud - d^2}{d+2u} = \frac{-d(d+u)}{(d+2u)} \quad \text{.... (2)}$$

$$\frac{1}{f_1} = \frac{d+2u}{ud} + \frac{1}{u} = 2\left(\frac{1}{d} + \frac{1}{u}\right) = \frac{2(d+u)}{du} \quad \text{.... (3)}$$

$$\frac{1}{f_2} = \frac{-1}{u+d} + \frac{d+2u}{d(d+u)}$$

$$\frac{1}{f_2} = \frac{-d+d+2u}{d(d+u)} = \frac{2u}{d(d+u)} \quad \text{... (4)}$$

$$(3) \times (4) \Rightarrow \frac{1}{f_1f_2} = \frac{4}{d^2}$$

$$d = 2\sqrt{f_1f_2}$$

$$\frac{1}{f_1} = \frac{2(d+u)}{du}$$

$$du = 2f_1d + 2uf_1$$

$$u = \frac{2f_1d}{d-2f_1} = \frac{2f_1 \times 2\sqrt{f_1f_2}}{2\sqrt{f_1f_2} - 2f_1}$$

$$= \frac{2f_1\sqrt{f_2}}{\sqrt{f_2} - \sqrt{f_1}}$$

$$d = 2\sqrt{f_af_b},\ u = \frac{2f_a\sqrt{f_b}}{\sqrt{f_b} - \sqrt{f_a}}$$

$$\frac{u}{d} = 1$$

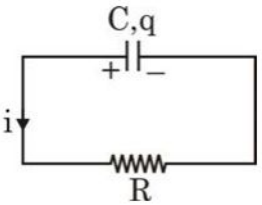
6. Ans (22)

$$10 = 6A + B$$

$$4 = 0.06A + B$$

$$\Rightarrow B \approx 4,\ A \approx 1$$

$$\therefore\ R = V + 4$$



$$i = \frac{-dq}{dt}$$

$$iR = \frac{q}{C} = V$$

$$\Rightarrow \frac{-dq}{dt}\left(\frac{q}{C} + 4\right) = \frac{q}{C}$$

$$\Rightarrow \int_{q_0}^{q_0/100} -dq \cdot \frac{(q+4)}{q} = \int_0^t dt \quad (q_0 = 6C)$$

$$\Rightarrow t = -\left[\frac{q_0}{100} - q_0 + 4\ell n\left(\frac{1}{100}\right)\right]$$

$$= \frac{99q_0}{100} + 4\ell n100 = \frac{99 \times 6}{100} + 8 \times 2$$

$$= 16 + 5.94 = 21.94 \approx 22 \text{ s}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A)

Λ_m of HCl > KCl > LiCl > CH_3COOH

2. Ans (B)

Kucherov reaction

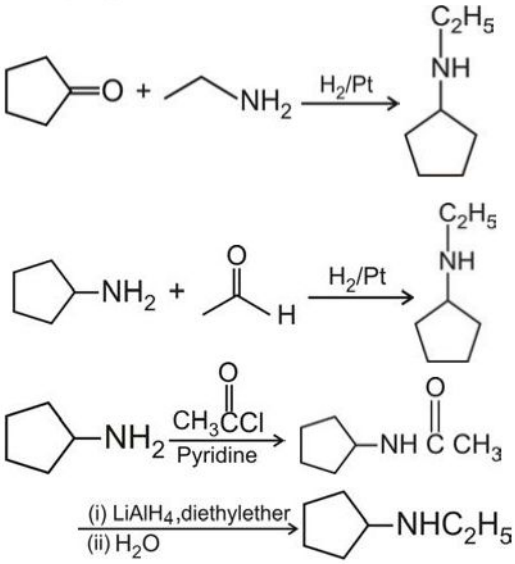
3. Ans (D)

$$Cr^{+3} \xrightarrow{OH^-/NaBO_3} CrO_4^{2-}$$

$$Mn^{+2} \xrightarrow{H^+/BiO_3^-} MnO_4^-$$

4. Ans (D)

Reaction (a, b, c) all give N-ethylcyclopentylamine as major product.



PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,C)

$$P_{total} = \frac{P_A}{Y_A} = \frac{P_B}{Y_B}$$

$$\frac{X_A P_A^o}{Y_A} = \frac{(1 - X_A) P_B^o}{(1 - Y_A)}$$

$$\frac{1}{X_A} = \frac{1}{Y_A} \times \frac{P_A^o}{P_B^o} + \frac{P_B^o - P_A^o}{P_B^o}$$

$$\text{Slope} = \frac{P_A^o}{P_B^o}, \text{ Intercept} = \frac{P_B^o - P_A^o}{P_B^o}$$

7. Ans (A,C)

$Cu^{+2} = [Ar]3d^9$ ↿⇂ 1 / ↿⇂ ↿ ↿ — unsymmetrical distribution in e_g orbitals cause Jahn Teller distortion

$Mn^{+3} = [Ar]3d^4$ 1 — / 1 1 1 — unsymmetrical distribution in e_g orbital cause Jahn Teller distortion

PART-2 : CHEMISTRY

SECTION-I (iii)

8. Ans (C)

Intermolecular force of attraction α a

Size of molecule α b

Boyle's temperature $(T_B) = \frac{a}{Rb}$

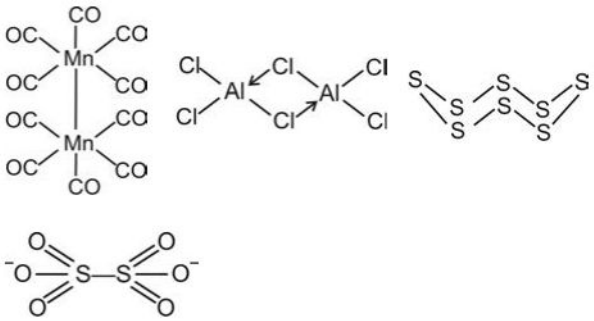
Critical temperature $(T_C) = 8 \times T_B$

9. Ans (A)

Strongly basic nucleophiles (II, III) cannot be used in presence of protic solvents (P,T)

Neutral Nucleophiles (IV) work better in presence of less protic solvents or aprotic solvents

10. Ans (B)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (4)

1, 3, 5, 7 statement are correct.

2. Ans (160)

$C(s) + O_2(g) \longrightarrow CO(g) + CO_2(g)$

a mole b mole

POAC on carbon, $1 \times 0.1 = a + b$

POAC on oxygen, $2 \times 0.08 = a + 2b$

$a = 0.04$ & $b = 0.06$

$5CO(g) + I_2O_5(s) \longrightarrow 5CO_2(g) + I_2$

0.04/5 mole

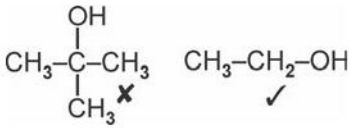
gmequivalent of I_2 = gmequivalent of hypo

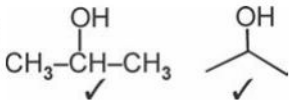
$0.008 \times 2 = 0.1 \times x \times 10^{-3}$

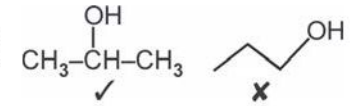
$x = 160$ ml

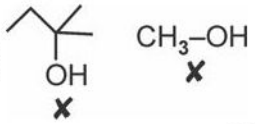
3. Ans (3)

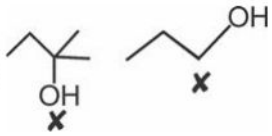
(I, III, VI)

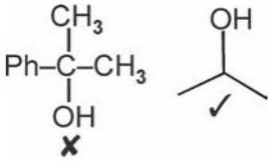
(1) 

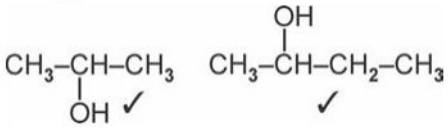
(2) 

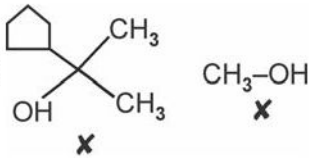
(3) 

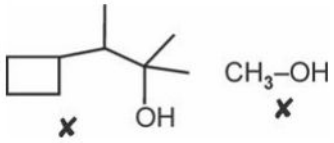
(4) 

(5) 

(6) 

(7) 

(8) 

(9) 

4. Ans (331)

X = KBr

P = HBr

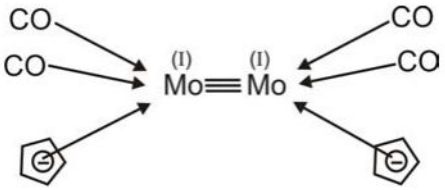
Q = Br_2

S = $Pb(CH_3COO)_2$

R = $PbBr_2$

Y = 2,4,6 tribromophenol

5. Ans (3)

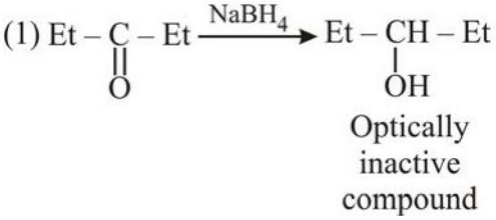


In above strength both Mo have 18 valence electrons

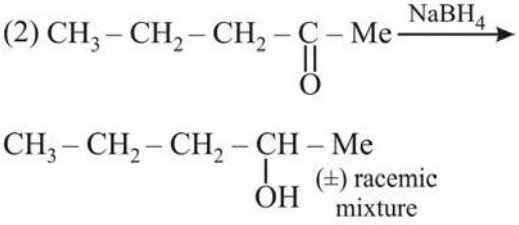
6. Ans (2)

M. wt 86 of ketone

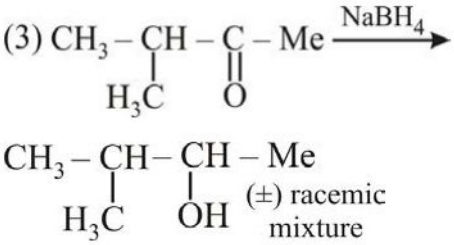
So m. formula = $C_5H_{10}O$

(1) 

Optically inactive compound

(2) 

(±) racemic mixture

(3) 

(±) racemic mixture

(2 ; 3)

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$\text{Let } I = \int_{\frac{1}{4}}^{4} \frac{\tan^{-1}x}{x}dx \quad ....(1)$$

$$\text{put } x = \frac{1}{t} \Rightarrow dx = -\frac{1}{t^2}dt$$

$$\Rightarrow I = \int_{\frac{1}{4}}^{4} \frac{\cot^{-1}x}{x}dx \quad ....(2)$$

$$\therefore (1) + (2) \Rightarrow 2I = \frac{\pi}{2}\int_{\frac{1}{4}}^{4} \frac{dx}{x}$$

$$\Rightarrow I = \frac{\pi}{4}\left(\ell n4 - \ell n\frac{1}{4}\right) \Rightarrow I = \frac{\pi}{2}\ell n4$$

$$= \pi \ \ell n\, 2$$

$$= 3.14 \times 0.69$$

$$= 2.166$$

$$\therefore 2[I] = 4$$

2. Ans (C)

$$\text{Put } x^2 = t \Rightarrow \int_0^1 f(t)dt = 1$$

$$\because \int_0^1 f(x)\,dx + \int_{f(0)}^{f(1)} f^{-1}(x)\,dx = 1.f(1) - 0.f(0)$$

$$\Rightarrow 1 + \int_1^2 f^{-1}(x)\,dx = 2$$

$$\Rightarrow \int_1^2 f^{-1}(x)dx = 1$$

3. Ans (C)

$$\text{Let } \tan\theta = t \Rightarrow \theta \in [0, \pi/2) \cup \{\pi, 2\pi\}$$

$$\Rightarrow \sqrt{t} = 2\sin\theta$$

squaring

$$\left(1 + \frac{1}{t^2}\right)t = 4$$

$$t + \frac{1}{t} = 4 \Rightarrow t = 2 + \sqrt{3}, 2 - \sqrt{3}$$

$$\Rightarrow \theta = 0, \frac{\pi}{12}, \frac{5\pi}{12}, \pi, 2\pi$$

4. Ans (C)

$$I = \lim_{n\to\infty}\int_{0^+}^{\pi/2} n\left(1 - (\sin x)^{\frac{1}{n}}\right)dx$$

$$= \int_{0^+}^{\pi/2}\left(\lim_{k\to 0}\frac{1-(\sin x)^k}{k}\right)dx$$

$$= \int_{0^+}^{\pi/2}\left(\lim_{k\to 0}\frac{-(\sin x)^k \ell n(\sin x)}{1}\right)dx$$

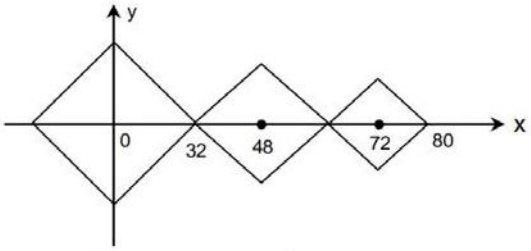
$$= -\int_{0^+}^{\pi/2} \ell n(\sin x)\,dx$$

$$= \frac{\pi}{2}\ell n2$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)



$a_2 = a_1 + \frac{3}{2} b_1 = 48,\ b_2 = \frac{b_1}{2} = 16$

$a_3 = 48 + \frac{3}{2} \times 16 = 72,\ b_3 = \frac{16}{2} = 8$

So, the three loops from i = 1 to i = 3 are alike

Now, area of ith loop (square) = $\frac{1}{2}$ (diagonal)^2

$A_i = \frac{1}{2}(2b_i)^2 = 2b_i^2$

So, the areas form a G.P series

$\therefore$ sum of the G.P up to infinite term is

$$\frac{A_i}{1-r} = 2(32)^2 \times 1 \frac{1}{1-\frac{1}{4}} = \frac{8}{3}(32)^2 \text{ sq. units}$$

6. Ans (A,B,C,D)

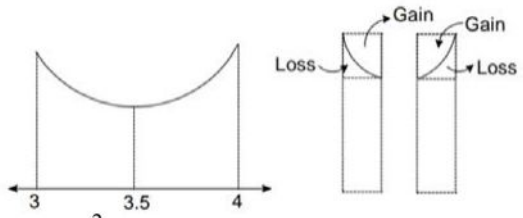
$\because y = mx - 2m - m^3$

$\Rightarrow 6 = 9m - 2m - m^3$

$\Rightarrow m^3 - 7m + 6 = 0 \Rightarrow m = 1, +2, -3$

$\therefore$ Centre is $\left(\frac{11}{2}, \frac{3}{2}\right)$

7. Ans (A,C)



$f(x) = x^2 - 7x + 13$

In both cases gain > loss

Curve is symmetrical about line $x = \frac{7}{2}$

$$\Rightarrow P > \int_3^4 f(x)\,dx \ \&\ Q > \int_3^4 f(x)\,dx$$

PART-3 : MATHEMATICS

SECTION-I (iii)

8. Ans (D)

(P) Given $a, d \in I^+$ and $a + 2d = 8 \Rightarrow a = 8 - 2d$

also, $191 < \frac{15}{2}(2a + 14d) < 196$

$\Rightarrow \frac{191}{15} < a + 7d < \frac{196}{15}$

$\Rightarrow \frac{191}{15} < 8 + 5d < \frac{196}{15} \Rightarrow \frac{71}{75} < d < \frac{76}{75}$ $\because d \in I^+$

$\therefore d = 1$ and $a = 6$

(Q) $(2\cos^2 y - 1) - (2\cos^2 x - 1) = 3a - 2$

$\cos^2 y - \cos^2 x = \frac{3a-2}{2}$ ...(1)

$\cos^2 x + \sin^2 y = \frac{a^2}{2}$ ...(2)

equation (1) + (2)

$1 = \frac{a^2 + 3a - 2}{2} \Rightarrow a^2 + 3a - 4 = 0$

$a = 1$ & $a = -4$ (impossible)

(R) $\sin^{-1} \frac{2x}{1+x^2} = \frac{\pi}{2} + \cos^{-1} x$

solution will exist if $\cos^{-1} x = 0$

& $\sin^{-1} \frac{2x}{1+x^2} = \frac{\pi}{2}$

which is possible only when x = 1.

(S) $\frac{1}{\Delta}\sqrt{abc(a+b+c)} = \frac{1}{\Delta}\sqrt{4R.\Delta.2S}$

$= \sqrt{\frac{8RS}{\Delta}} = \sqrt{\frac{8R}{r}} \geqslant \sqrt{8.2} = 4$

$\because r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$\frac{r}{R} = 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \Rightarrow \frac{R}{r} \geq 2.$

In any triangle $\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} \leqslant \frac{1}{8}$

9. Ans (B)

(P) $\int_{-2}^{6} f(\{x\})dx = \int_{-2}^{2} f(\{x\})dx + \int_{2}^{6} f(\{x\})dx$

$\therefore \int_{2}^{6} f(\{x\})dx = -2$

(Q) $\left|\vec{a}+\vec{b}\right|^2 + \left|\vec{a}-\vec{b}\right|^2 = 2\left(|\vec{a}|^2 + \left|\vec{b}\right|^2\right)$

$\Rightarrow |\vec{a}| = 5$

(R) $2\int_{0}^{\pi} |\text{sgn}(\tan x) + \text{sgn}(\cot x)|\, dx$

$2\left(\int_{0}^{\pi/2} 2dx + \int_{\pi/2}^{\pi} 2dx\right) = 8.\frac{\pi}{2} = 4\pi$

(S) $A = \begin{bmatrix} 1 & \left(\frac{1}{2} + \frac{1}{2^2} + \ldots\right) \\ 0 & 1 \end{bmatrix}$

Sum of all elements of matrix A = 3

10. Ans (A)

(P) Required probability $= \frac{{}^{n-3}C_4}{{}^{n}C_4}$

(Q) $P_2 = \frac{n-3}{n} \times \frac{3}{(n-1)}$ or

$\left(\frac{{}^{n-3}C_1}{{}^{n}C_1} \cdot \frac{{}^{3}C_1}{{}^{n-1}C_1}\right)$

(R) $P_{n-2} = \frac{{}^{n-3}C_{n-3}}{{}^{n}C_{n-3}} = \frac{1}{{}^{n}C_3}$

(S) $P_r = \frac{{}^{n-3}C_{r-1}}{{}^{n}C_{r-1}} \cdot \frac{3}{(n+1-r)}$ or

$\left(P_r = \frac{{}^{n-3}C_{r-1}}{{}^{n}C_{r-1}} \cdot \frac{{}^{3}C_1}{{}^{n-r+1}C_1}\right)$

11. Ans (B)

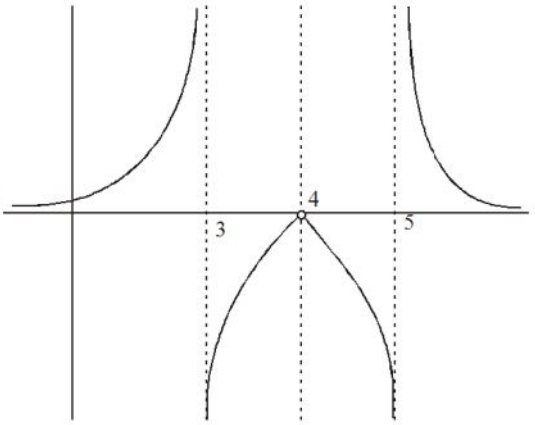
(P) $\lim_{x\to 0}\left[\text{min.}\left((y+1)^2+3\right).\frac{\sin x}{x}\right]$

$= 2$

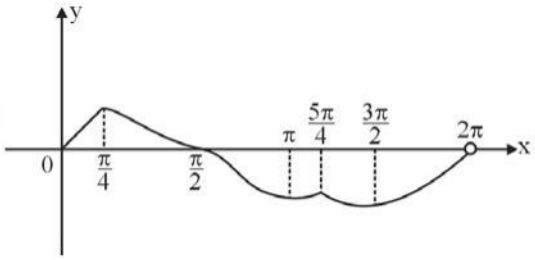
(Q) $\lim_{x\to 0} \frac{\sin^{-1}\left(\frac{2x}{1+x^2}\right)}{\left(\frac{2x}{1+x^2}\right)} \cdot \frac{2x}{x.(1+x^2)}$

$= 1 . \frac{2}{1+0} = 2$

(R)



(S)



PART-3 : MATHEMATICS

SECTION-II

1. Ans (75)

$1 + {}^{4}C_1 \times 2 + \frac{\lfloor 4}{\lfloor 2 \lfloor 2 \lfloor 2} \times \lfloor 2 + \frac{\lfloor 4 \times \lfloor 3}{\lfloor 2 \lfloor 1 \lfloor 1 \lfloor 2} + \lfloor 4 = 75$

or

$1 + {}^{4}C_3 \lfloor 2 + {}^{4}C_2 \lfloor 3 + \lfloor 4$

2. Ans (3)

By properties PA = PB = $\sqrt{10}$

$$\frac{x-1}{-1/\sqrt{10}} = \frac{y-5}{3/\sqrt{10}} = \pm\sqrt{10}$$

$\Rightarrow A \equiv (0, 8); B \equiv (2, 2)$

Slopes of asymptotes are –7, and 1

If angle between asymptotes be θ then

$$\tan\theta = \left|\frac{-7-1}{1-7}\right| = \frac{4}{3}$$

Now, $2\tan^{-1}\frac{b}{a} = \tan^{-1}\frac{4}{3} \Rightarrow \frac{\frac{2b}{a}}{1-\frac{b^2}{a^2}} = \frac{4}{3}$

$\Rightarrow \frac{b}{a} = \frac{1}{2}$ $\left(\frac{b}{a} = 2 \text{ rejected}\right)$ ...(1)

$\frac{b^2}{a^2} = e^2 - 1 \Rightarrow e = \frac{\sqrt{5}}{2}$

Also, area of $\Delta CAB = ab$

$$ab = \left|\frac{1}{2}[1(8-2) + 0(2-1) + 2(1-8)]\right| = 4 \quad ...(2)$$

$b^2 = 2$, $a = 2b = 2\sqrt{2}$ length of latus rectum

$$= \frac{2b^2}{a} = \frac{4}{2\sqrt{2}} = \sqrt{2} = \ell$$

$\therefore [e^2 + \ell^2] = 3$

3. Ans (1)

point of intersection of lines at origin O = (0, 0, 0),

$$A = (3,2,2) \text{ \& } B = \left(\frac{8}{3}, 2, \frac{4}{3}\right)$$

$$\text{Area of triangle} = \frac{1}{2}\left|\overrightarrow{OA} \times \overrightarrow{OB}\right|$$

$$\text{Area of triangle} = \frac{1}{2}\left|-\frac{4}{3}\hat{i} + \frac{4}{3}\hat{j} + \frac{2}{3}\hat{k}\right| = 1$$

4. Ans (6)

$A^{-1}B^2 + AB = 0$

$\Rightarrow A^{-1}B.BB^{-1} + AB.B^{-1} = 0$

$\Rightarrow AA^{-1}B + A.A = 0$

$\Rightarrow B = -A^2$ .....(1)

& $|A| = 5$ .....(2)

$A^6 - 2A^4B + A^2B^2 = 4A^6$

$\therefore |4A^6| = 4^3 |A|^6 . = 2^6 . 5^6 = 10^6$

$\therefore n = 6$

5. Ans (1)

$$(z+1)\begin{vmatrix} z+\omega^2 & 1 \\ 1 & z+\omega \end{vmatrix} + \omega\begin{vmatrix} 1 & \omega \\ z+\omega & \omega^2 \end{vmatrix} + \omega^2\begin{vmatrix} \omega & z+\omega^2 \\ \omega^2 & 1 \end{vmatrix} = 0$$

As $1 + \omega + \omega^2 = 0$, $\omega^4 = \omega$

$\Rightarrow (z + 1)(z^2 + z(\omega + \omega^2) + \omega^3 - 1) +$

$\omega(\omega^2 - z\omega - \omega^2) + \omega^2(\omega - z\omega^2 - \omega^4) = 0$

$\Rightarrow (z + 1)(z^2 - z) + z(-\omega^2 - \omega) = 0$

$\Rightarrow z^3 - z^2 + z^2 - z + z = 0$

$\Rightarrow z^3 = 0$

$\Rightarrow z = 0$ is only 1 complex number

6. Ans (0)

Replacing x by 1 + x in (1), we get

$f'(1 + x) = -f'(1 - x)$

Let $I = \int_{-1}^{1} f'(1+x)x^2e^{x^2}dx$ .....(2)

$I = \int_{-1}^{1} f'(1-x).x^2e^{x^2}dx$ (using king)

$I = -\int_{-1}^{1} f'(1+x).x^2e^{x^2}dx$

$(\because f'(1 + x) = -f'(1 - x))$

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)
AIOT
21-04-2024

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY — PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	B	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,B	B,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	3.00	3.12 to 3.16	0.80	2.80 to 2.81		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	16	9	3	8	114

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	D	A	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	B,C	B,C,D	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.60	42.50	8.00	7.10		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	300	5	5	3	7	2

PART-3 : MATHEMATICS

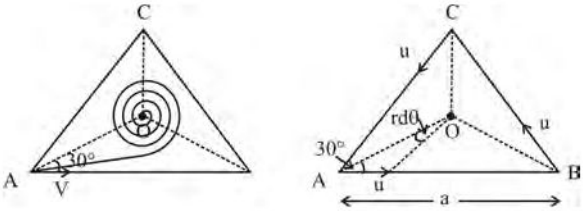
SECTION-I (i)	Q.	1	2	3	4		
	A.	C	D	A	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,C,D	B,D	B,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	2.00	9.00	67.00	22.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	0	80	4	4	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)



$$\frac{dr}{dt} = -v\cos 30^\circ = -\frac{\sqrt{3}}{2}V$$

$$r\frac{d\theta}{dt} = v\sin 30^\circ = \frac{v}{2}$$

$$\frac{1}{r}\frac{d\theta}{dr} = -\sqrt{3}$$

$$\int_{r_0}^{r}\frac{dr}{r} = -\sqrt{3}\int_{r_0}^{r} d\theta \Rightarrow r = r_0 e^{-\sqrt{3}\theta}$$

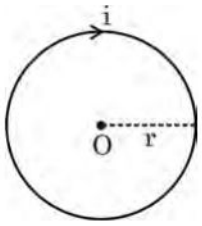
When A completes one revolution $\theta = 2\pi$

$$\text{Time taken } t = \frac{r_0\left(1 - e^{-2\sqrt{3}\pi}\right)}{\sqrt{3}v/2}$$

$$\text{Distance travelled } D = vt = \frac{2r_0}{\sqrt{3}}\left(1 - e^{-2\sqrt{3}\pi}\right)$$

$$D = \frac{2a}{3}\left(1 - e^{-2\sqrt{3}\pi}\right)$$

2. Ans (C)



$$B_0 = \frac{\mu_0 i}{2\pi r}$$

$$\phi = B_0\pi r^2$$

$$\varepsilon = \frac{\Delta\phi}{\Delta t} = \frac{\mu_0}{2\pi r}\pi r^2\frac{\Delta i}{\Delta t} = \frac{\mu_0 r}{2}\frac{\Delta i}{\Delta t}$$

$$\text{Res.} = \frac{\varepsilon}{i} \Rightarrow \frac{\rho\ell}{A} = \frac{\varepsilon}{i} \Rightarrow \rho = \frac{\varepsilon A}{\ell i}$$

$$\rho = \frac{\mu_o r}{2}\frac{\Delta i}{\Delta t} \times \frac{A}{2\pi r} \times \frac{1}{i}$$

$$\rho = \frac{\mu_o}{4\pi} \times \frac{\Delta i}{\Delta t} \times A \times \frac{1}{i}$$

$r = 5$ cm

$$A = \frac{\pi d^2}{4},\ d = 1\text{mm}$$

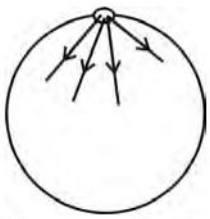
$$\frac{\Delta i}{\Delta t} = \frac{10^{-6}}{365 \times 86400}\frac{A}{5}$$

$i = 100$ A

$$\rho = \frac{10^{-7} \times 3.17 \times 10^{-14} \times 0.78 \times 10^{-6}}{10^2}$$

$$\rho = 2.5 \times 10^{-29}$$

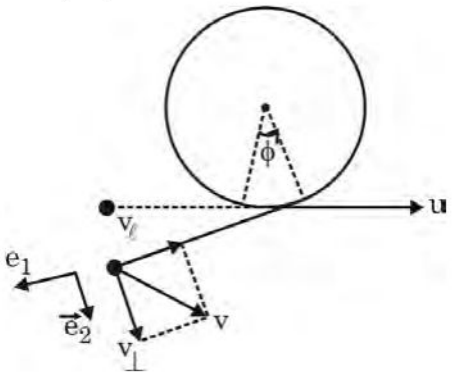
3. Ans (B)



Power received $= I_0\pi r^2$

$$\text{Intensity} = \frac{I_0\pi r^2}{4\pi R^2} = \frac{I_0 r^2}{4R^2}$$

4. Ans (C)



$\vec{e}_1$ and $\vec{e}_2$ are unit vectors along and perpendicular to the thread.

$$\vec{v} = -v_\ell \vec{e}_1 + v_\perp \vec{e}_2 \quad .....(1)$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -u\frac{d\vec{e}_1}{dt} - \underbrace{\frac{du}{dt}\vec{e}_1}_{0} + v_\perp \frac{d\vec{e}_2}{dt} + \frac{dv_\perp}{dt}\vec{e}_2$$

$$\vec{a} = -u\frac{d\vec{e}_1}{dt} + v_\perp \frac{d\vec{e}_2}{dt} + \frac{dv_\perp}{dt}\vec{e}_2$$

vector $\vec{e}_1$ and $\vec{e}_2$ rotates as a rigid body with angular velocity

$$\vec{\omega} = \frac{d\phi}{dt}(\vec{e}_1 \times \vec{e}_2)$$

$$\frac{d\vec{e}_1}{dt} = \vec{\omega} \times \vec{e}_1 = \frac{d\phi}{dt}\vec{e}_2$$

$$\frac{d\vec{e}_2}{dt} = \vec{\omega} \times \vec{e}_2 = -\frac{d\phi}{dt}\vec{e}_1$$

$$\vec{a} = -v_\perp \frac{d\phi}{dt}\vec{e}_1 + \left(-u\frac{d\phi}{dt} + \frac{dv_\perp}{dt}\right)\vec{e}_1$$

row there is no force is direction $\vec{e}_2$

$$\Rightarrow \frac{dv_\perp}{dt} = u\frac{d\vec{\phi}}{dt} \Rightarrow v\perp = u\phi$$

When P is about to detach

$$\phi = 2N\pi$$

$$v\perp = 2N\pi u$$

$$v = \sqrt{v_\ell^2 + v_\perp^2} = u\sqrt{1 + (2N\pi)^2}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C,D)

Frequency of surface MN $\left(f'\right) = \frac{10f}{7}$

$$\lambda_A = \frac{7v}{f'} = \frac{49v}{10f}$$

$$\lambda_B = \frac{7v}{f'} = \frac{49v}{10f}$$

$$f_A = \left(\frac{7v - \frac{v}{6}}{7v}\right) f' = \left(\frac{205}{147}\right) f$$

$$f_B = \left(\frac{7v + \frac{v}{9}}{7v}\right) f' = \left(\frac{640}{441}\right) f$$

6. Ans (A,B)

Let ω_0 be the angular velocity of the rod just before the collision. Then from conservation of mechanical energy:

$$\frac{1}{2}\frac{ML^2}{3}\omega_0^2 = Mg\frac{L}{2}(1 - \sin\theta)$$

When $\sin\theta = \frac{L}{2L} = \frac{1}{2}$

$$\Rightarrow \omega_0 = \sqrt{\frac{3g}{2L}}$$

because of hinge reaction, linear momentum of the system cannot be conserved in the horizontal direction.

But angular momentum of the system can be conserved about the hinge, just during the collision, neglecting the effect of gravity and its torque about the hinge during the collision time because gravity is non-impulsive force.

If ω = angular velocity of the (Rod + Bullet) system just after collision,

Then $\left(\frac{ML^2}{3}+mL^2\right)\omega = \frac{ML^2}{3}\omega_0 + Mv_0\frac{L}{2}$

$\frac{1}{2}\left(\frac{ML^2}{3}+mL^2\right)\omega^2 = Mg\frac{L}{4}+mg\frac{L}{2}$

putting $M = 3m$: $\omega^2 L = \frac{5g}{4}$ ....(iii)

using (i), (ii) and (iii)

$v_0 = \sqrt{gL}\left(\sqrt{6}+\sqrt{20}\right)$

(b) After collision, when the system makes an angle 60° with the horizontal, total torque about the hinge $= Mg = \frac{L}{2}$

$\cos 60^o + mg\, L \cos 60^o = \frac{5}{4}mgL,$

by putting $M = 3m$

Moment of inertia of the system about the hinge $= \frac{ML^2}{3} + mL^2 = 2mL^2$

By putting $M = 3m$

Hence, angular acceleration of the system

$= \frac{\text{Torque}}{\text{Moment of inertia}} = \frac{(5/4)\,mgL}{2mL^2} = \frac{5g}{8L}$

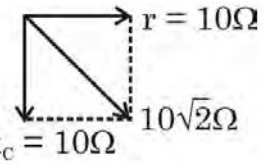
Therefore, tangential acceleration of the midpoint

$= \left(\frac{L}{2}\right)\left(\frac{5g}{8L}\right) = \frac{5g}{16}$

7. Ans (B,C)

(1) In the source

$x_C = \frac{1}{\omega C} = 10\Omega,\ r = 10\Omega$



$z_1 = r - x_C j$

(2) Let load has $z_2 = R + xj$

z of existing ckt $\Rightarrow z_1 = r + (-x_C)j$

$z_{eq} = z_1 + z_2 \Rightarrow z_{eq} = (R + r) + (x - x_C)j$

$P_{load} = i_{rms\ load}\, v_{rms\ load} \cos\phi_{load}$

$P_{load} = \frac{\varepsilon_{rms}^2}{z_{eq}^2}R$

$P_{load} = \frac{\varepsilon_{rms}^2 R}{\sqrt{(R+r)^2 + (x - x_C)^2}}$

for P_{load} to be maximum

$x = x_C = 10\Omega$

$P_{load} = \frac{\varepsilon_{rms}^2 R}{(R+r)^2}$

$\frac{dP_{load}}{dR} = 0$

$\Rightarrow R = r$

$\Rightarrow (P_{load})_{max} = \frac{\varepsilon_{rms}^2}{4R}$

Now $R = r = 10\Omega$

$\varepsilon_{rms} = \frac{15}{\sqrt{2}}$volt

$(P_{load})_{max} = \frac{15 \times 15}{2 \times 4 \times 10}$

$(P_{load})_{max} = \frac{45}{16}$watt

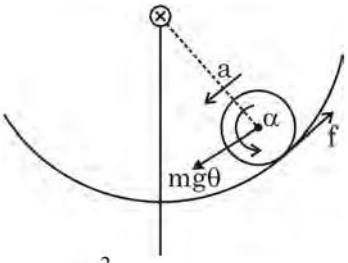
$z_2 = z_{load} = 10 + 10j$

P.f. $= \phi = \frac{10}{10\sqrt{2}} = \frac{1}{\sqrt{2}}$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (3.12 to 3.16)

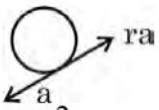


$fr = \frac{mr^2}{2}\alpha \rightarrow f = \frac{mr\alpha}{2}$ .....(1)

$mg\theta - f = ma$

$mg\theta - \frac{ma}{2} = ma$

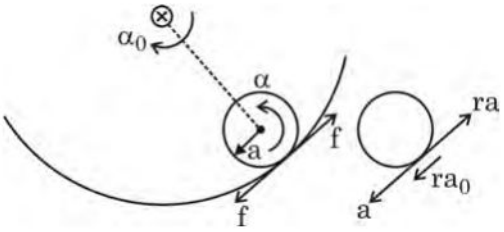
$mg\theta = \frac{3}{2}ma$



$a = \frac{2g}{3}\theta \rightarrow \theta = \frac{2g}{3(R-r)}\theta$

$\omega = \sqrt{\frac{2g}{3(R-r)}} = \sqrt{\frac{2\times 1000}{3\times 0.75}} = \sqrt{\frac{16\times 5}{9}}$

$= \frac{4}{3}\times 2.25 = \frac{4}{3}\times\frac{225}{100} = 3$



$a - r\alpha = R\alpha_0$ .....(iii)

$fr = \frac{mr^2}{2}\alpha \rightarrow f = \frac{mr\alpha}{2}$

$mg\theta - f = ma$ .....(4)

$fR = MR^2\alpha_0$

$f = MR\alpha_0$

Put from (iii)

$f = M(a - r\alpha)$ .....(iv)

by (i) & (iv)

$\frac{mr\alpha}{2} = Ma - Mr\alpha$

$\left(\frac{m}{2} + M\right) r\alpha = Ma$

$\gamma\alpha = \frac{M}{\left(\frac{m}{2} + M\right)}a$

Put in equation (ii)

$mg\theta - \frac{m}{2}\left(\frac{2m}{m+2M}\right)a = ma$

$g\theta = a + \left(\frac{M}{m+2M}\right)a$

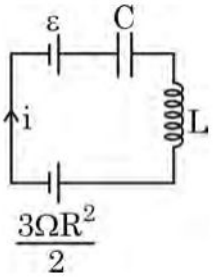
$\theta = \frac{g(m+2M)}{(m+3m)(R-r)}\theta$

$\omega = \sqrt{\left(\frac{m+2M}{m+3M}\right)\frac{g}{(R-r)}}$

Given M = 1kg, m = 1kg

$\omega = \sqrt{10}$

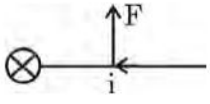
3. Ans (0.80)



$\varepsilon - \frac{q}{C} - L\frac{di}{dt} - \frac{3R^2}{2}\Omega = 0$

$\frac{d\varepsilon}{dt} - \frac{1}{C}\frac{dq}{dt} - L\frac{d^2i}{dt^2} - \frac{3R^2}{2}\frac{d\Omega}{dt} = 0$

$-\frac{1}{C}i - L\frac{d^2i}{dt^2} - \frac{BR^2}{2}\alpha = 0$ .....(1)



$\tau = iRB\frac{R}{2} = \frac{iR^2B}{2}$

$$mR^2\alpha = \frac{iR^2B}{2}$$

$$\alpha = \frac{iB}{2m}$$

$$-\frac{i}{C} - L\frac{d^2i}{dt^2} - \frac{BR^2}{2} \times \frac{iB}{2m} = 0$$

$$\frac{i}{C} + L\frac{d^2i}{dt^2} + \frac{B^2R^2}{4m}i = 0$$

$$\frac{d^2i}{dt^2} = \underbrace{-\left(\frac{1}{LC} + \frac{B^2R^2}{4mL}\right)}_{\omega^2} i$$

$$i = i_0 \sin(\omega t + \delta)$$

where $\omega = \sqrt{\frac{1}{LC} + \frac{B^2R^2}{4mL}}$

as at $t = 0$, $i = 0 \Rightarrow \delta = 0$

and $t = 0$, $\varepsilon = \ell\frac{di}{dt}$

$$\frac{di}{dt} = i_0\omega\cos\omega t$$

$$\varepsilon = Li_0\omega\cos\omega t = Li_0\omega$$

$$i_0 = \frac{\varepsilon}{L\omega}$$

$$i = \frac{\varepsilon}{L\omega}\sin\omega t$$

$$\alpha = \frac{B}{2m}\frac{\varepsilon}{1\omega}\sin\omega t$$

$$\Omega = \frac{B\varepsilon}{2mL\omega^2}[\cos\omega t]_0^t$$

$$\Omega = \frac{B\varepsilon}{2mL\omega^2}(1 - \cos\omega t)$$

$$\Omega_{max} = \frac{2B\varepsilon}{2mL\omega^2} = \frac{B\varepsilon}{mL\omega^2}$$

and when $\cos\omega t = 1$

$$\omega t = \pi \Rightarrow t = \frac{\pi}{\omega}$$

now $\omega = \sqrt{\frac{1}{1} + \frac{1}{4}} = \frac{\sqrt{5}}{2}$

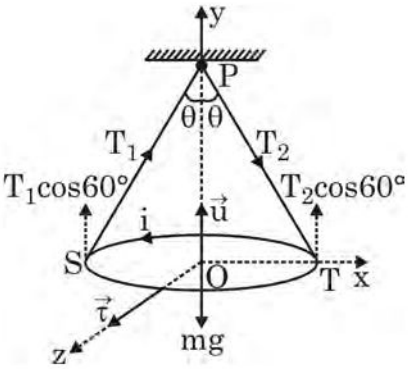
$$\omega_{max} = \frac{1 \times 1 \times 4}{1 \times 1 \times 5} \Rightarrow \Omega_{max} = 0.8 \text{ rad/s}$$

$$t = \frac{3.14 \times 2}{2.24} = 2.80 \text{ sec}$$

PART-1 : PHYSICS

SECTION-II (ii)

5. Ans (3)



$T_1\cos60° + T_2\cos60° = mg$

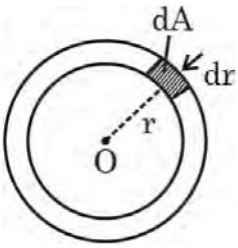
$T_1 + T_2 = 2mg$ ... (i)

Taking torque about 'O'

$(T_2\cos60°)\,R + \left|\vec{\mu} \times \vec{B}\right| = (T_1\cos60°)\,R$ .... (ii)

On solving $\frac{T_1}{T_2} = 3$

6. Ans (16)



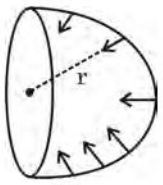
$$dF = \left(\frac{4\pi G\rho r}{3}\right) dm$$

$$dF = \frac{4\pi G\rho r}{3} \times \rho dr dA$$

$$\frac{dF}{dA} = \frac{4\pi G\rho^2}{3} dr \Rightarrow \text{Gravitational pressure}$$

$$P = \frac{4\pi G\rho^2 r}{3} dr$$

Now force on hemisphere $F = \int P\,dA$



$$F = \int_{r=0}^{r=A} \frac{4\pi G\rho^2}{3} dr\pi r^2$$

$$F = \frac{4\pi^2}{3} G\rho^2 \int_0^R r^3 dr = \frac{4\pi^2 G\rho^2}{12} R^4$$

$$F = \frac{4\pi^2 G}{12} \frac{M^2}{\left(\frac{4}{3}\pi R^3\right)^2} R^4 \Rightarrow F = \frac{4\pi^2 G}{12} \frac{M^2}{\frac{16}{9}} \frac{R^4}{\pi^2 R^C R^2}$$

$$F = \frac{3}{16} \frac{GM^2}{R^2}$$

7. Ans (9)

The radioactive decay series is given

$$A \xrightarrow{T_{1/2} = 30 \text{ days}} B \xrightarrow{T_{1/2} = 45 \text{ days}} C$$

Initially $N_A(0) : N_B(0) = 3 : 1$

$$\frac{dN_A}{dt} + \lambda_A N_A = 0$$

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N_B$$

$$\frac{dN_B}{dt} = \lambda_A N_A - \lambda_B N$$

$$\frac{dN_C}{dt} = \lambda_B N_B$$

Initially $N_A = N_A(0) e^{-\lambda A t}$

$$N_B = c_1 e^{-\lambda_B t} + \frac{\lambda_A N_A(0) e^{-\lambda_A t}}{-\lambda_A + \lambda_B}$$

If $N_A(0) = \frac{3}{4} N_o$, $N_B(0) = \frac{1}{4} N_o$

then we get, $c_1 = \frac{5}{2} N_o$

$\therefore$ $N_A(t) = \frac{3}{4} N_o \left(\frac{1}{2}\right)^{\frac{t}{T_{\frac{1}{2}}}} = \frac{3}{4} N_o \left(\frac{1}{2}\right)^{\frac{t}{30 \text{ days}}}$ and,

$$N_B(t) = \left[\frac{5}{2} N_o \left(\frac{1}{2}\right)^{\frac{t}{45 \text{ days}}} - \frac{9}{4} N_o \left(\frac{1}{2}\right)^{\frac{t}{30 \text{ days}}}\right]$$

Now, $\frac{\lambda_A N_A}{\lambda_B N_B} = \frac{9}{22}$ i.e. $\frac{N_A}{N_B} = \frac{3}{11}$

or, $\left(\frac{1}{2}\right)^{\frac{-t}{90}} = 2$

or, $t = 90$ days

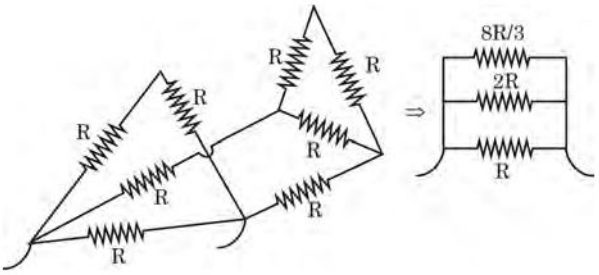
8. Ans (3)

$$\sigma\pi R^2 T_0^4 = \sigma 2\pi R^2 T^4 + \sigma\pi R^2 T^4$$

$$T_0^4 = 3T^4 \Rightarrow T = \left(\frac{1}{3}\right)^{1/4} T_0$$

$$T = 10\,(3)^{3/4}$$

9. Ans (8)



$$\frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{3}{8R}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{8+4+3}{8R} = \frac{15}{8R}$$

$$R_{eq} = \frac{8R}{15}$$

10. Ans (114)

$$\tan i = \frac{4}{3} \quad \therefore \; i = 53^\circ$$

$\therefore r = 37$

$$\tan i_2 = \frac{3/2}{4/3} = \frac{9}{8}$$

$\therefore$ $i_2 = 48.4$

$\alpha + 90 + r + 90 - i_2 = 180$

$\alpha = i_2 - 37 = (11.4)^\circ$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B)

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

Slope $= -\Delta S^\circ$

2. Ans (D)

$$\text{P.E.} = \frac{kq_1q_2}{r} = -\frac{1}{4\pi\varepsilon_0} \times \frac{3e \times e}{a_0 \times \frac{(2)^2}{3}}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,C)

$Cr_2O_7^{2-} + 6e^- + 14H^+ \longrightarrow 2Cr^{3+} + 7H_2O$

$$\text{At pH} = 3,\ E = 1.33 - \frac{0.06}{6}\log\frac{1}{(10^{-3})^{14}} = 0.91\ V$$

$$\text{At pH} = 0,\ E = 1.33 - \frac{0.06}{6}\log\frac{1}{(1)^{14}} = 1.33\ V.$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (2.60)

Rate law = $k^1\,[CHCl_3]\,[Cl_2]^{1/2}$

$$k^1 = k_2\left(\frac{k_1}{k_{-1}}\right)^{1/2} = 1.3 \times 10^{-2} \times \left(\frac{4.8\times10^3}{1.2\times10^3}\right)^{1/2}$$

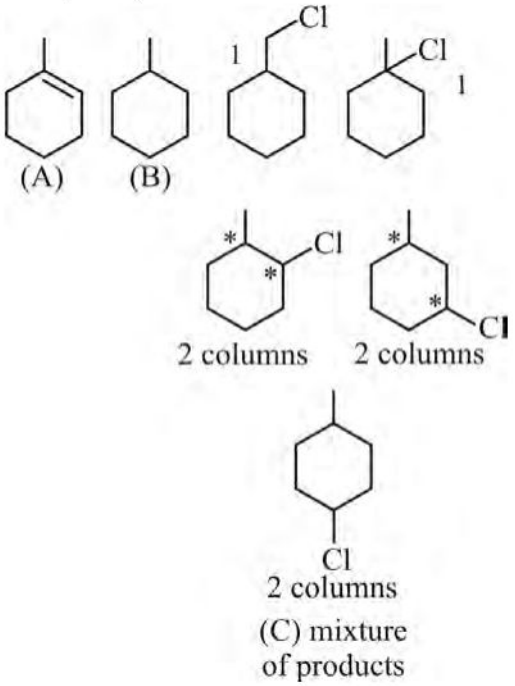
$= 2.6 \times 10^{-2}\ M^{-1/2}\ sec^{-1}$.

2. Ans (42.50)

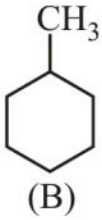
$$E_{a_{overall}} = E_{a_2} + \frac{1}{2}\left(E_{a_1} - E_{a_{-1}}\right)$$

$= 45 + 0.5\ (10 - 15) = 42.5$ kJ/mol.

3. Ans (8.00)



4. Ans (7.10)



M = 98 gm $\quad n_B = \frac{9.8}{98} = 0.1$

$n_B = n_{Cl_2} = 0.1$

$W_{Cl_2} = 0.1 \times 71 = 7.10$ gm

PART-2 : CHEMISTRY

SECTION-II (ii)

5. Ans (300)

$K_{SP} = [M^{y+}]^x\,[SO_4^{2-}]^y$

$\log K_{SP} = x \log [M^{y+}] + y \log [SO_4^{2-}]$

From plot

$$\log [M^{y+}] = \frac{\log K_{SP}}{x} + \frac{y}{x}\log [SO_4^{2-}]$$

$$\frac{y}{x} = \frac{2}{4} = \frac{1}{2}$$

$x = 2$ and $y = 1$

$K_{SP} = [10^{-2}]^2 \times [10^{-4}] = 10^{-8}\ M^3$

$$z = \frac{x+y}{K_{SP}\ \text{of}\ M_x(SO_4)_y} = \frac{2+1}{10^{-8}} = 300 \times 10^6$$

10. Ans (2)

v, viii are basic amino acids, which exist as cation at pH = 7

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

$$\text{Let } A = \lim_{n\to\infty} \frac{\left[\left(n^2+1^2\right)\left(n^2+2^2\right)\ldots\left(n^2+n^2\right)\right]^{\frac{m}{n}}}{n^{2m}}$$

$$\Rightarrow \ln A = \lim_{n\to\infty} \frac{m}{n}\sum_{r=1}^{n} \ln\left(n^2+r^2\right) - 2m\ell nn$$

$$= \lim_{n\to\infty} m\left[\frac{\sum_{r=1}^{n}\ln\left(n^2+r^2\right) - 2n\ln n}{n}\right]$$

$$\Rightarrow \ln A = m\left[\ln 2 + \frac{\pi}{2} - 2\right]$$

$$\Rightarrow A = e^{m\left[\ln 2+\frac{\pi}{2}-2\right]}$$

$$= \left[2.e^{\frac{\pi}{2}}.e^{-2}\right]^m$$

$$A = \left[\frac{2\sqrt{e^{\pi}}}{e^2}\right]^m \quad \therefore a = 2$$

2. Ans (D)

$$\sum_{k=0}^{1012}\left[{}^{1012}C_k \sum_{r=k}^{2024}\frac{{}^{r}C_k.{}^{2024}C_r}{{}^{2024}C_k}\right]$$

$$= \sum_{k=0}^{1012} {}^{1012}C_k.2^{2024-k}$$

$$= 2^{2024}\sum_{k=1}^{1012} {}^{1012}C_k.\left(\frac{1}{2}\right)^k$$

$$= 2^{1012}\cdot 3^{1012}$$

$$= 6^{1012}$$

$$= (5+1)^{1012}$$

$$= 25\lambda + 5061$$

$$= 25\mu + 11$$

Remainder = 11

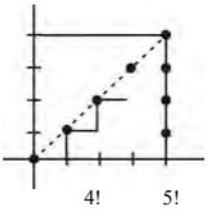
3. Ans (A)

(1) 3 move diagonally

(2) 2 move diagonally, 1 left, 1 down

(3) 1 move diagonally, 2 left, 2 move down

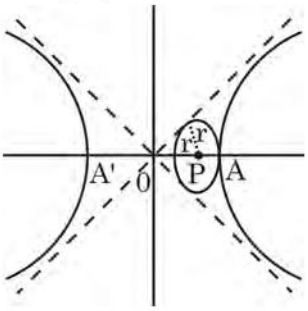
(4) 3 move left, 3 move down



$$\frac{1}{3^3} + \frac{\frac{4!}{2!}}{3^4} + \frac{\frac{5!}{2!2!}}{3^5} + \frac{\frac{6!}{3!3!}}{3^6} = \frac{245}{3^6}$$

$$(1,1) \text{ to } (0,0) \longrightarrow \frac{1}{3}.\frac{245}{3^6} \Rightarrow \frac{245}{3^7} \Rightarrow \frac{x}{3^y}$$

4. Ans (B)



$$OP = r\,\text{cosec}\theta$$

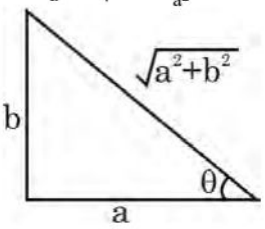
$$OA = a = r + r\,\text{cosec}\theta$$

$$\Rightarrow r = \frac{a}{1+\cos ec\theta}$$

$$= \frac{a}{1+\frac{\sqrt{a^2+b^2}}{b}}$$

$$= \frac{b}{\frac{b}{a}+\sqrt{\frac{a^2+b^2}{a^2}}}$$

$$= \frac{b}{\frac{b}{a}+\sqrt{1+\frac{b^2}{a^2}}}$$



$$r = \frac{b}{\sqrt{e^2-1}+e} = \frac{b\left(e-\sqrt{e^2-1}\right)}{e^2-(e^2-1)}$$

$$\Rightarrow \frac{r}{b} = e - \sqrt{e^2-1}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B,C,D)

$3AABB + 3AB = 2ABAB + 2AB$

$\Rightarrow 3A(AB + I)B = 2A(BA + I)B$

$\Rightarrow 27|A||AB + I||B| = 8|A||BA + I||B|$

as $|I + AB| = |I + BA| \neq 0$

we get $|B| = 0$

6. Ans (B,D)

$$f(x) = \lim_{n\to\infty} \frac{\cot x + 1}{\cot x - 1} \times \frac{(\cot x)^{2^{n+1}} - 1}{(\cot x)^{2^{n+1}} + 1}$$

$$\forall\, x \in \left(0, \frac{\pi}{2}\right] - \left\{\frac{\pi}{4}\right\}$$

$$\therefore f(x) = \frac{\cot x + 1}{\cot x - 1}, x \in \left(0, \frac{\pi}{4}\right)$$

$$= \frac{\cot x + 1}{1 - \cot x}, x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

and $f\left(\frac{\pi}{4}\right) = 2, f(0) = 0$

$$\text{Now } f'(x) = \frac{2}{(\cot x - 1)^2}.\cos ec^2 x, x \in \left(0, \frac{\pi}{4}\right)$$

$$= \frac{-1}{(\cot x - 1)^2}\cos ec^2 x, x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$\therefore$ f(x) is increasing in $\left(0, \frac{\pi}{4}\right)$ and f(x) is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$.

Also f(x) has three local minima at $x = 0$, $x = \frac{\pi}{4}, x = \frac{\pi}{2}$ and f(x) does not have local maxima

7. Ans (B,D)

$z = e^{i\pi/13} \Rightarrow z^{13} = -1$

$$z^{13} + 1 = 0 \Rightarrow \frac{1 + z^{13}}{1 + z} = 0$$

$$\Rightarrow 1 - z + z^2 - z^3 + ... + z^{12} = 0 \quad ...(1)$$

$1 - z^{13} = 2$

$$\Rightarrow \frac{1 - z^{13}}{1 - z} = \frac{2}{1 - z}$$

$$\Rightarrow 1 + z + z^2 + z^3 + ... + z^{12} = \frac{2}{1 - z} \quad ...(2)$$

from (1) & (2)

$$1 + z^2 + z^4 + z^6 + z^8 + z^{12} = \frac{1}{1 - z} \text{ (B option)}$$

$$\& \; z + z^3 + z^5 + ... + z^{11} = \frac{1}{1 - z} \text{ (D option)}$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.00)

Let the reflection of $2x + y + z - 5 = 0$ about

$x - y + 2z + 5 = 0$

be plane $(2x + y + z - 5) + k(x - y + 2z + 5) = 0$

$$\text{Now, } \left|\frac{(2 + k) - (1 - k) + 2(1 + 2k)}{\sqrt{(2 + k)^2 + (1 - k)^2 + (1 + 2k)^2}.\sqrt{6}}\right| = \left|\frac{2 - 1 + 2}{\sqrt{6}.\sqrt{6}}\right|$$

$\Rightarrow 6k + 3 = 3\sqrt{k^2 + k + 1}$

$\Rightarrow k = -1$ OR $k = 0$ (reject)

$\therefore$ resultant plane is $x + 2y - z - 10 = 0$

So λ is equal to minimum distance of

$x + 2y - z - 10 = 0$ from any point on $x^2 + y^2 = 5$

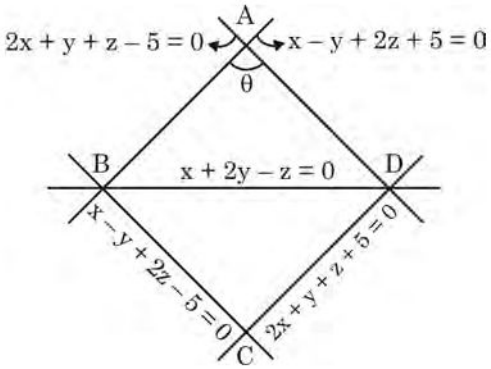
Let a point on circle be

$\left(\sqrt{5}\cos\theta, \sqrt{5}\sin\theta, 0\right)$

$$\text{Then } \perp \text{ distance} = \left|\frac{\sqrt{5}\cos\theta + 2\sqrt{5}\sin\theta - 10}{\sqrt{6}}\right|$$

$$\therefore \lambda = \frac{5}{\sqrt{6}} \Rightarrow [\lambda] = 2$$

2. Ans (9.00)



Perpendicular distance between opposite sides of ABCD is $\frac{10}{\sqrt{6}}$

$\therefore$ Area of cross section $= \frac{1}{2} \times \frac{10}{\sqrt{6}} \times \frac{10}{\sqrt{6}} \times \operatorname{cosec}\theta$

$= \frac{1}{2} \times \frac{10}{\sqrt{6}} \times \frac{10}{\sqrt{6}} \times \frac{2}{\sqrt{3}}$

$= \frac{50}{3\sqrt{3}}$

$\therefore [A] = 9$

3. Ans (67.00)

Consider $I = \int_0^1 (1-x)^m x^n dx$,, where $m, n \in N$.

Put $x = \sin^2\theta \Rightarrow dx = 2\sin\theta \cos\theta \, d\theta$

$\Rightarrow I = 2\int_0^{\frac{\pi}{2}} (\cos\theta)^{2m+1} (\sin\theta)^{2n+1} d\theta$

$\Rightarrow I = \frac{2 \times (2m)(2m-2) \times \ldots \times 2 \times (2n)(2n-2) \times \ldots \times 2}{(2m+2n+2)(2m+2n) \times \ldots \times 2}$

$\Rightarrow I = \frac{m!n!}{(m+n+1)!}$

$\therefore I_1 = \frac{6!4!}{11!}, I_2 = \frac{5!3!}{9!}$

$\Rightarrow \frac{I_1}{I_2} = \frac{6 \times 4}{11 \times 10} = \frac{12}{55}$

4. Ans (22.00)

$I = \frac{1}{2^5} \int_0^{\frac{\pi}{4}} \frac{(\cos x - \sin x)^{10}}{(\cos x)^{10}} \times \frac{(\sin x)^7}{(\cos x)^7} . \sec^2 x \, dx$

$\tan x = t$

$I = \frac{1}{2^5} \int_0^1 (1-t)^{10} . t^7 dt$

$I = \frac{10!7!}{2^5 \times 18!}$

PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (3)

$(2xy - y)\left(x^4 + x^2 + 2\right) = 0$

$\Rightarrow y = 0; x = \frac{1}{2}$

Also, $x^3 + y^4 = x$

$y = 0; x = -1, 0, 1$

$x = \frac{1}{2}; y = \pm\left(\frac{3}{8}\right)^{1/4}$

6. Ans (0)

$(2x + 1)^2 + (2y - 1)^2 = 8086$

$x, y \in I \quad 8k_1 + 1 + 8k_2 + 1 = 8086$

$8(k_1 + k_2) = 8084$

not possible

7. Ans (80)

Let $x_1 = 3$.

$3(x_2x_3 + x_4x_5 + x_2x_5) + x_3x_4 (x_2 + x_5)$

$x_2 + x_5$ must be 3λ

i.e., (1, 2), (1, 5), (2, 4), (4, 5) $\longrightarrow$ 8 ways

Req. ways $= 5 \times 8 \times 2 = 80$

8. Ans (4)

At $x = 0$ $\quad \frac{-k}{(-2)^4} + \frac{6}{(-1)^4} = 0 \Rightarrow k = 96$

$(\sin\theta + 2\cos\theta)(\sin\theta + \cos\theta) = 0$

$\tan\theta = -1, -2$

9. Ans (4)

$A^2 = 7A$

$\Rightarrow \text{tr}(A^n) = 0^n + 7^n = 7^n$

i.e., $n = 4$

10. Ans (5)

$\frac{x^2}{100} + \frac{y^2}{225} = 1; (x-2)^2 + (y-3)^2 = 25$

$(y-3)^2 = 6(x-7)$

